Household Search and the Equilibrium Gender Pay Gap*

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Abstract

We develop and calibrate an equilibrium labor market model to quantify the respective roles of labor supply and demand in generating gender gaps in employment, hours and wages over the life cycle. Households with and without children make joint labor market decisions, while firms offer wages anticipating the households' labor supply responses. Calibrating the model to German household data, we find that labor supply factors account for approximately 60 percent of the gender earnings gap, primarily reflecting gender differences in preferences for home hours. Firm-side forces are also quantitatively important: statistical discrimination and firms' gender wedges account for the remaining 40 percent. Our model further reveals that women sort into less productive firms, especially after childbirth, and that statistical discrimination emanates almost exclusively from the presence of joint taxation and intra-household risk sharing. Counterfactual reforms reducing parental-leave duration and the parental allowance increase fathers' leave take-up, but have only modest and short-lived effects on gender gaps.

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1 Introduction

Despite significant convergence over the last decades, women still have substantially lower employment rates, working hours and earnings than men in the developed world. Recent literature has identified parenthood as the main driver of the persistence in earnings differentials across genders (Kleven et al., 2019a,b; Adda et al., 2017). The birth of a child impacts the choices of parents in the labor market by permanently increasing both the financial need of the household and the demand for hours spent at home. To meet these needs, parents may decide to specialize in either paid work or childcare activities (Becker, 1985), which may amplify pre-existing earnings differentials: fathers of young children tend to work more hours and earn higher wages, while mothers often leave the labor force or work part-time for a prolonged time. Importantly, firms anticipate the career choices made at the household level and may decide to offer different jobs and wages to men and women.[1]

So far, there is limited literature on the role of demand and supply forces in shaping labor market outcomes for men and women at different stages of the life cycle, and in particular before and after childbirth. Yet, quantifying these forces is key for identifying effective policies aimed at reducing gender differences in labor market outcomes. Many developed countries enacted family-friendly programs to facilitate the transition of workers into parenthood, such as parental leave legislation or parental allowances. However, these programs appear to have limited or negligible beneficial effects on labor market outcomes (Albanesi et al., 2023; Kleven et al., 2024). Furthermore, recent research indicates that firms adjust their hiring and pay policies in response to family policies (Ginja et al., 2023; Fernández-Kranz and Rodríguez-Planas, 2021; Huebener et al., 2025). Therefore, it is essential to untangle the mechanisms behind both worker and employer decisions to design effective policies.

This paper quantifies the driving forces of gender gaps over the life cycle, taking into account hiring and wage decisions of firms, and job acceptance, job quitting, and parental leave take-up decisions of workers in single and couple households. We consider a structural equilibrium model in which firms differ in productivity, create full-time and part-time jobs, and offer wages to workers of either gender. Male and female workers live in single or couple households and undergo several life-cycle stages characterized by the presence and the age of children. Men and women differ in their preferences for home hours, in job search effort, and in their initial human capital when entering the labor market. Human capital is accumulated during employment spells, and depreciates in periods of leave and unemployment.

[1] The idea that firms statistically discriminate in their hiring and pay policies based on perceived differences in working hours across genders also shows up in Albanesi and Olivetti (2009) and Fernández-Kranz and Rodríguez-Planas (2021).

The output of a worker-firm match is increasing in firm productivity, working hours, and in the human capital of the worker. Firms have gender-specific preferences for hiring men or women into full-time or part-time jobs, which we model as gender wedges that distort the perceived productivity of a worker-job match, reflecting either taste-based discrimination on the side of managers or a different task assignment of men and women within the firm. Firms may also statistically discriminate against women when their skill distribution or their job search and quitting behavior (into parental leave, unemployment, or other jobs) differ from those of men. The matching process between workers and firms is frictional. Both unemployed and employed workers match with firms that offer wages to attract and retain their workers. We build on Burdett and Mortensen (1998), Bontemps et al. (2000) and Fang and Shephard (2019).

We calibrate the model using the German household panel data to match, among others, labor market participation, working hours, and wages of men and women at different life-cycle stages. Differences in job search effort across genders, household types and life-cycle stages are calibrated to the average number of applications reported in our data. The calibrated model replicates gender gaps in wages and hours, the sorting of men and women across firms of different sizes, as well as the child penalty in the labor market. We use the model to quantify the relevance of the driving forces of gender gaps in employment, hours and wages: worker-side effects (initial skills, preferences for home hours, and job search effort), firms' gender wedges, and statistical discrimination. Equalizing all structural parameters between men and women fully eliminates all gender differences in our model, while activating some of the differences allows us to measure their respective contributions to the gender gaps.

We find that labor supply forces account for approximately 60 percent of the total gender earnings gap over the life cycle. This component is driven primarily by gender differences in preferences for home hours; heterogeneity in job search intensity matters less, while differences in the skills at labor market entry contribute only a negligible amount. Demand-side forces are quantitatively important: statistical discrimination explains 19 percent of the gap, while gender wedges account for 21 percent. These forces contribute very differently to gender gaps over the life cycle. While the demand-side gap generated by gender wedges is roughly constant across life-cycle stages, the pronounced widening of the gaps in employment, working hours and earnings around childbirth is driven almost entirely by supply-side forces, in particular gender differences in preferences for home hours and in job search.

We further decompose the gender gap in hourly wages into a between-firm component, a within-firm component, a component due to differences in accumulated skills, and a part-time wage penalty. On average, the within-firm component is the largest contributor to

the hourly wage gap but is fairly constant over the life cycle. Around childbirth, however, the wage gap widens, and this widening is driven mainly by the between-firm component, as mothers increasingly sort into less-productive lower-paying firms, together with the part-time penalty. The skill gap widens too, but is quantitatively less important.

A central mechanism operates through household specialization and its feedback on firms' wage setting. In our model, household income is pooled in two ways: spouses are taxed jointly under a progressive schedule, and they share consumption through intra-household insurance. This income pooling shapes the labor supply of primary and secondary earners. In the absence of income pooling, both male and female labor supply becomes less elastic, to which firms respond by offering lower wages, especially to men in full-time jobs. When we counterfactually remove income pooling, such that both spouses make separate labor market decisions, the gender earnings gap falls substantially, from about 50 percent to 32 percent. Strikingly, statistical discrimination vanishes almost entirely once income is no longer pooled, so that statistical discrimination in our model arises as a consequence of joint taxation and intra-household risk sharing.

Finally, we use the calibrated model to evaluate counterfactual reforms of two central family policies, parental leave and parental allowance. In the benchmark economy, as in our German data, mothers take up parental leave far more often and for much longer than fathers. Take-up also depends on a worker's own pre-birth working hours and on the spouse's labor market state. While cutting the parental allowance has only small effects on parental leave take-up, halving the maximum leave duration raises fathers' take-up from 16 to 41 percent and shortens mothers' return to work; yet, the resulting effects on the gender gaps in employment, hours and wages remain modest and short-lived.

Our paper contributes to four strands of the literature. First, we contribute to the literature on joint household search initiated by Dey and Flinn (2008) and Guler et al. (2012), and further developed by Flabbi and Mabli (2018), García-Pérez and Rendón (2020), Pilossoph and Wee (2021), and Wang (2019). Most closely related, Pilossoph and Wee (2021) show that income pooling raises the reservation wages of both spouses relative to singles; building on this insight, we establish that income pooling makes the labor supply of couple workers more wage-elastic, a feature that shapes the gender wage gap once firms set wages endogenously. Importantly, existing work on labor market dynamics with joint household search leaves the labor demand side (job creation and wage setting) exogenous. One notable exception is Fang and Shephard (2019) who structurally estimate an equilibrium joint search model, yet they do not analyze gender gaps and abstract from differences in working hours.

Second, we build on the literature that studies life-cycle labor supply choices of couples

in structural models, such as Adda et al. (2017), Borella et al. (2018) and Guner et al. (2012). A closely related strand examines how the tax-and-transfer system shapes the labor supply of married women: Bick and Fuchs-Schündeln (2018) show that the joint taxation of couples accounts for much of the cross-country variation in married women's hours, while Guner et al. (2020) and Hannusch (2025) study how child-related transfers and family policies affect maternal employment. This literature takes wages and the labor demand side as given. Our paper contributes by embedding joint household labor supply in an equilibrium model in which firms make hiring and wage-offer decisions in the presence of labor market frictions, so that the joint taxation and income pooling of couples feed back into firms' wage setting.

Third, we contribute to a recent literature examining the gender earnings gap using structural models with labor market frictions. Faberman et al. (2026), Morchio and Moser (2026) and Xiao (2024) build equilibrium models in which men and women search for jobs created by firms. These jobs differ in productivity and amenities, while Xiao (2024) also takes childbirth events and parental leave into account. Different from our work, these papers neither consider joint career choices of couples, nor do they allow for the prevalence of part-time work which is a key determinant for gender gaps in many countries (cf. Kleven et al., 2019a). The same applies to Amano-Patino et al. (2025) who consider an equilibrium model with on-the-job search of both genders which additionally includes human capital accumulation on the job. While these papers feature gendered monopsony in firms' wage setting, we trace the gender differences in wage markdowns to household income pooling: because pooling raises couples' reservation wages and makes their labor supply more wage-elastic, especially for higher-earning men, firms optimally apply smaller markdowns to men, which our model identifies as a quantitatively important source of the gender wage gap.

Fourth, our evaluation of parental-leave reforms relates to theoretical and empirical research. A structural literature studies parental-leave policies in quantitative models, including Erosa et al. (2010) in a general-equilibrium matching model, Albrecht et al. (2024) with a focus on social norms, and Kim and Yum (2025) on the joint effects on fertility and labor supply. Most closely related to our work is Almar et al. (2025), who develop a quantitative life cycle model with household-firm interactions and evaluate parental-leave policies, but with a focus on firms' promotion and training decisions rather than on wage-setting, working hours, and statistical discrimination. A largely empirical literature evaluates the German parental leave system to which our model is calibrated: Schönberg and Ludsteck (2014) and Frodermann et al. (2023) estimate effects on mothers' employment and earnings, while Huebener et al. (2025) study how firms absorb parental-leave absences.

Our paper combines the key ingredients of these strands of the literature, namely joint

household decisions about working hours over the life cycle, labor market frictions, and endogenous labor demand and wage setting. Our results highlight that the combination of these ingredients is important for understanding gender earnings gaps: First, women and men, especially after childbirth, sort differentially into firms along the wage-productivity ladder. Second, statistical discrimination arises in response to joint household taxation and intra-household risk sharing. Third, parental leave take-up and the return to jobs differ by the spousal employment state and working hours.

The rest of the paper is organized as follows. Section 2 presents the model, Section 3 describes the calibration strategy and the model fit, and Section 4 presents our results. Section 5 concludes.

2 The Model

We consider a random search job-posting model with heterogeneous workers and firms which is based on Burdett and Mortensen (1998) and Bontemps et al. (2000). There are men and women workers who live in single and couple households that transit through family stages over the life cycle. Firms offer jobs that differ in working hours and wages. Workers make job acceptance and job quitting decisions, and their skills appreciate (depreciate) while they are employed (non-employed). Under parental leave regulations, parents with young children have the right to leave their job for a limited time after which they are allowed to return to the job. The model is set up in continuous time and we consider a stationary equilibrium.

2.1 The Environment

Workers and Households

We consider men and women workers indexed by $g = m, f$ who are members of either single or couple households without transitions between the two. Single households are indexed by the gender of the household head, while couple households include two working-age members of opposite gender. All households transit through five family states indexed by $k = 1, \ldots, 5$, where $k = 5$ is retirement and $k < 5$ are working-age households: $k = 1$ denotes a childless fertile household, $k = 2$ ($k = 3$) denotes a household with young (old) children and $k = 4$ is a childless infertile household. The transition rate from state k to state k' is denoted $\Phi^g_{kk'}$ for a single household of gender g and $\Phi_{kk'}$ for a couple household. Retirement is an absorbing state. To keep the working-age population constant, every retiring single or couple household is replaced by a new single or couple household in family state $k = 1$ with non-employed

workers.

Household utility in retirement is normalized to zero. During working age, the utility flow of a single household of gender $g = m, f$ is

$$u_g(c, h_g; k) = \ln c + \psi_k^g \frac{(\bar{h} - h_g)^{1-\gamma}}{1-\gamma} + \alpha_k^g \mathbf{1}_{\{h_g > 0\}} ,$$

while the utility flow of a couple household is

$$u(c, h_m, h_f; k) = \sum_{g=m,f} u_g(c_g, h_g; k) ,$$

where $c_m = c_f = c/2$ and c is household consumption, h_g is working hours of gender $g \in \{m, f\}$. The utility parameters ψ_k^g and α_k^g govern labor supply at the extensive and intensive margin. These parameters are allowed to vary by gender and household type, while the hours endowment $\bar{h}$ and the utility curvature parameter γ are assumed not to vary with demographics. Households do not save so that consumption is equal to household labor income after taxes and transfers. Households discount future utility at rate ρ.

Workers differ in skills $s \in \{s^1, \ldots, s^n\} \subset \mathbb{R}_{++}$ where $s^i < s^{i+1}$. Skills may either appreciate or depreciate, depending on a worker's current employment state and working hours. For an employed worker, the skill goes up to $s^+(s^i) \equiv s^{i+1}$ at transition rate π_x^+ (if $s^i < s^n$). For a non-employed worker, the skill goes down to $s^-(s^i) \equiv s^{i-1}$ with probability π_x^- (if $s^i > s^1$). Here x is the worker's state vector, which is formally defined below, and which includes working hours, so that skill accumulation may differ between full-time or part-time jobs. New households entering family state $k = 1$ draw initial skills with probability $\zeta^g(s)$ for singles of gender g and with probability $\zeta(s_m, s_f)$ for couple households.

Firms

There is a fixed unit mass of firms that differ in productivity y which is distributed with cumulative distribution function $\Gamma(.)$. Firms offer full-time and part-time jobs with hours requirements $h \in \{h^F, h^P\}$. They operate linear production technologies that sum the output of all jobs. If a job in a firm of productivity y is filled with a worker of skill s working h hours, the output is ysh. That is, worker skills and firm productivity are complementary such that ys is the hourly productivity of the match.

To allow for the possibility that firms treat male and female job applicants differently, we introduce a (perceived) productivity gap between male and female workers that may vary with working hours. These gaps may reflect a gendered task assignment of firms or a

productivity misperception (taste-based discrimination) on the side of managers. Specifically, the perceived output of a female worker in a full-time (part-time) job is multiplied by the gender-wedge parameter η^F (η^P). The gender wedge may reflect a productivity discount (when $\eta < 1$) or a productivity premium (when $\eta > 1$) of a female worker.

When a firm aiming to fill a job with hours requirement h meets a male (female) worker, it offers an hourly wage per efficiency unit which is denoted w_m (w_f). Specifically, we allow firms to offer different wages to male or female job applicants at the matching stage.[2] Firms maximize the profit value of their job offers, taking into account workers' labor supply choices as reflected in job acceptance, job quitting, parental leave and hours reductions after parental leave. Therefore, gender differences in wage offers reflect not only the gender wedges introduced above, but also the firms' rational expectations about worker choices regarding quits, parental leave take-up and hours reductions (statistical discrimination). We write Δ^h for the fraction of jobs created with hours requirement h, and $F^{g,h}(w)$ for the distribution of hourly skill adjusted wage offers to workers of gender g in these jobs. These objects are determined in equilibrium.

Parental Leave, Right to Part-Time, Taxes and Transfers

In line with parental leave legislation in many countries, parents of young children have the right to leave their job for a limited time period during which the employer is required to hold the job. In our model, parental leave ends at transition rate τ which is a policy parameter. Parental leave also ends automatically when the children grow older, i.e. when the household leaves family state $k = 2$. In both cases, the worker decides to either return to the previous job or quit the job. We allow the worker to enter parental leave either upon receiving the young child or, in a couple household with young children, when the partner's parental leave spell ends. The latter reflects parental leave legislation that grants this right to both parents.

We also allow parents to request an hours reduction of their full-time job, i.e. a switch from full-time to part-time working hours (at the same hourly wage), which aligns with part-time worker rights in many European countries. To keep the model tractable, we assume those part-time transitions to be permanent, i.e. the employer is not required to provide a full-time job later on. We also restrict such requests to be made upon childbirth (transition

[2] In line with anti-discrimination legislation in modern societies, firms are not allowed to openly discriminate in their pay policies. However, private firms retain discretionary control over which worker to hire and which wage to offer. Our model reflects these hiring and pay choices and how they respond to the policy environment. It can also account for within-firm gender pay differences which are still sizable in many countries (e.g. Hennig and Stadler, 2023).

from $k = 1$ to $k = 2$) or at the end of a parental leave spell.[3]

The government maintains a tax-transfer system that determines how disposable household income depends on the labor earnings of the household's workers and on the family composition. Specifically, disposable income in single and couple households is given by $Y^s(x_g, k)$ and $Y^c(x_m, x_f, k)$, where $x_g \equiv (w_g, h_g, s_g, p_g)$ is the state vector of a worker of gender g. The state vector includes the current wage w_g, contractual working hours h_g (which are both zero when the worker is non-employed), worker skill s_g, and the indicator variable $p_g \in \{0, 1, 2\}$ which indicates parental leave or part-time right status. Specifically, we denote $p_g = 0$ if a worker is in regular employment, $p_g = 1$ for workers on parental leave, and $p_g = 2$ for workers who were hired into a full-time job (i.e. contractual hours are $h_g = h^F$) but who requested a permanent part-time reduction. Note that the tax-transfer schedules $Y^s(.)$ and $Y^c(.)$ depend on the family state k and on parental leave status, hence incorporates child allowance and parental allowance as further policy parameters that affect the labor market choices of parents.

Matching and Separations

We allow job-offer rates to vary with gender and with the family state in order to reflect how job search may differ between men and women with or without children. Non-employed workers of gender g in a single or couple household $\chi = s, c$ and family state k meet firms at flow rate $\lambda^n_{g,k,\chi}$, so that a worker is offered a job with hours requirement h at flow rate $\lambda^n_{g,k,\chi}\Delta^h$ with a wage offer w drawn from $F^{g,h}(.)$. The worker then decides whether to accept or reject the offer. Employed workers encounter outside offers at flow rates $\lambda^e_{g,k,\chi}$ whose hours requirements and wages are drawn from the same distribution. The employed worker then decides whether or not to accept the outside offer. Note that workers on parental leave are considered non-employed.

A worker in a job with hours requirement h separates from the job into non-employment at exogenous layoff rate δ_h. Any worker is allowed to voluntarily quit into non-employment when the family status changes, when parental leave ends, or in a couple household, when the partner accepts a new job or returns from parental leave.

[3]Germany introduced right-to-part-time legislation in the year 2001 which granted the right to work part-time to all workers (with or without children) employed in firms with more than 15 employees. Paule-Paludkiewicz (2024) finds that the legislation only had an effect on part-time work of mothers with young children, which justifies our assumption.

2.2 Household Decisions

We denote by $V(x_m, x_f, k)$ the value function of a couple where the state of a worker of gender g is $x_g = (w_g, h_g, s_g, p_g)$, as explained before, and k is the joint family state. With abuse of notation, we also write $V(x_g, x_{-g}, k)$ for generic gender g so that $-g$ is the gender of the partner. We further simplify notation and denote by $\xi_g \in \{e, n\}$ the employment status of worker g. The Bellman equation of the couple household is[4]

$$
\begin{aligned}
D(x_m, x_f, k)V(x_m, x_f, k) = u(Y^c(x_m, x_f, k), h_m, h_f) + \sum_{k'} \Phi_{kk'} \bar{V}(x_m, x_f, k') \qquad (2.1)\\
+ \sum_{g=m,f} \Big\{ \pi^+_{x_g} V(x^+_g, x_{-g}, k) + \pi^-_{x_g} V(x^-_g, x_{-g}, k)\\
+ \sum_{\hat{h}=h_F,h_P} \lambda^{e_g/n_g}_{g,k,c} \Delta^{\hat{h}} \int \max\left[V(x_g, x_{-g}, k), V(\hat{x}_g, x_{-g}, k), V(\hat{x}_g, x^n_{-g}, k)\right] dF^{g,\hat{h}}(\hat{w})\\
+ \delta_{h_g} V(x^n_g, x_{-g}, k) + \mathbb{I}_{p_g=1} \tau \bar{V}^\tau(x_g, x_{-g}, k) \Big\}
\end{aligned}
$$

where $D(x_m, x_f, k) \equiv \rho + \sum_{k'} \Phi_{kk'} + \sum_g \{\pi^+_{x_g} + \pi^-_{x_g} + \sum_{\hat{h}} \lambda^{\xi_g}_{g,k,c} \Delta^{\hat{h}} + \mathbb{I}_{p_g=0}\tau + \delta_{h_g}\}$. The first line includes the continuation utility after a change of the family state, denoted $\bar{V}$ and further described below. The second line of equation (2.1) indicates value changes following skill appreciation or depreciation, where we use shortened notation for the worker's continuation states $x^+ = (w, h, s^+(s), p)$ (skill appreciation) and $x^- = (w, h, s^-(s), p)$ (skill depreciation). The third line shows the continuation value if one partner receives a job offer where $\hat{x} = (\hat{w}, \hat{h}, s, 0)$ is the continuation state of the new job $(\hat{w}, \hat{h})$ which also erases any potential parental leave status of the previous job. If the job offer is accepted, the partner may quit to non-employment where we use $x^n = (0, 0, s, 0)$ to denote the continuation state of a newly non-employed worker with skill level s inherited from the previous state x. The fourth line shows the transitions after a job separation and after the end of a parental leave spell of worker g. The latter happens when $p_g = 1$ and at transition rate τ. In this case, continuation utility is denoted $\bar{V}^\tau$ and further described below.

Consider continuation utility after a change of the family state denoted $\bar{V}$. If that transition occurs from $k = 1$ into state $k' = 2$ (young children), either one of the two partners may take parental leave which dominates an outright quit or an hours reduction, as parental leave comes together with parental allowance payments and entails the option of returning to the previous job while still permitting the search for other jobs. A transition to

[4]The Bellman equation applies to all employment states of the two workers. To simplify notation, we set $\delta_{h=0} = 0$ because non-employed workers cannot separate from a job.

any other family state may induce a quit to non-employment for either of the two workers. Further, a transition from $k = 2$ into $k = 3$ (household with older children) may entail a return from parental leave (current state $x = (w, h, s, 1)$) to the previous job with the same hours, as indicated by the continuation state $x^r = (w, h, s, 0)$, or a return to the same job at lower hours (if the former job was a full time job $h = h_F$) with continuation state $x^l = (w, h, s, 2)$, and in both cases the partner may or may not quit the job. As any partner (potentially) has the option to return to the job (same or lower hours) or to quit the job with continuation state $x^n = (0, 0, s, 0)$, we generically denote the continuation state $x_g^{(r,l,n)}$, and the couple makes an optimal choice about these joint continuation states. Taken together, the continuation utility is

$$\bar{V}(x_g, x_{-g}, k) = \begin{cases} \max\left[V(x_g, x_{-g}, k), V(x_g^p, x_{-g}, k), V(x_g, x_{-g}^p, k)\right] & \text{if } k = 2 \ , \\ \max\left[V(x_g^{(r,l,n)}, x_{-g}^{(r,l,n)}, k)\right] & \text{if } k = 3 \ , \\ \max\left[V(x_g, x_{-g}, k), V(x_g^n, x_{-g}, k), V(x_g, x_{-g}^n, k)\right] & \text{if } k \neq 2, 3, K \ , \\ 0 & \text{if } k = K \ . \end{cases}$$

Finally, consider continuation utility after the end of parental leave of worker g denoted $\bar{V}^\tau(x_g, x_{-g}, k)$. Here worker g may either return to the previous job at the same hours with continuation state $x^r = (w, h, s, 0)$ or, if the previous job was a full-time job, return at lower hours with continuation state $x^l = (w, h, s, 2)$, or quit the job to non-employment with continuation state $x^n = (0, 0, s, 0)$. When the worker returns to the previous job (at same or lower hours), the partner may take parental leave where $x^p = (w, h, s, 1)$ is the state of a worker entering parental leave from state $x = (w, h, s, 0)$. Note that these transitions are all conditional on the household including young children ($k = 2$).[5] Consequently we have

$$\bar{V}^\tau(x_g, x_{-g}, k) = \max\left[V(x_g^r, x_{-g}, k), V(x_g^l, x_{-g}, k), V(x_g^r, x_{-g}^p, k), V(x_g^l, x_{-g}^p, k), V(x_g^n, x_{-g}, k)\right] \ .$$

All decisions about job acceptance, quitting, entering parental leave, or exercising the right to return to the previous job (at the same or reduced hours) involve binary policy functions where we use the notation

$$\Omega\left((x_g^0, x_{-g}^0); (x_g^i, x_{-g}^i)_{i=1}^I; k\right) = \begin{cases} 1 & \text{if } V(x_g^0, x_{-g}^0, k) > V(x_g^i, x_{-g}^i, k) \text{ for all } i = 1, \ldots, I \ , \\ 0 & \text{else} \ , \end{cases}$$

[5] Alternating, repeated spells of parental leave of the same worker are permitted in this model framework. They are also in line with German law as long as children are young and the total duration of parental leave does not exceed three years.

to indicate that (x_g^0, x_{-g}^0) is preferred over any of the alternatives (x_g^i, x_{-g}^i), $i > 0$. In the numerical solution of the model, we smooth out these discrete choices by adding small taste shocks which simply add noise to the household decisions.[6] In this modification, the above function Ω indicates *the proportion of households* that prefer (x_g^0, x_{-g}^0) over the stated alternatives. Our results are not sensitive to the inclusion of these taste shocks. Appendix B contains further details about this extension and it also presents the Bellman equation for single households with value and policy functions V^g, Ω^g.

2.3 Stationary Household Distribution

In a stationary equilibrium, let $G^g(x_g, k)$, $g = m, f$, and $G(x_m, x_f, k)$ denote the measures of single and couple households in the respective states (x_g, k), (x_m, x_f, k). Again with abuse of notation, we also write $G(x_g, x_{-g}, k)$ for couple households with generic gender g. We solve for the stationary measures by equating the inflows and outflows to/from a given household state.

Consider a couple household in state (x_m, x_f, k). The outflow from this state is equal to

$$\mathcal{O}(x_m, x_f, k) \equiv G(x_m, x_f, k) \Bigg[\underbrace{\sum_{k'} \Phi_{kk'}}_{\text{family state change}} + \sum_{g=m,f} \Big\{ \underbrace{\pi^+_{x_g} + \pi^-_{x_g}}_{g\text{'s skill adjustment}} + \underbrace{\delta_{h_g}}_{g \text{ is laid off}} + \underbrace{\sum_{\hat{h}} \lambda^{\xi_g}_{g,k,c} \Delta^{\hat{h}} \int \left[1 - \Omega((x_g, x_{-g}); (\hat{x}_g, x_{-g}), (\hat{x}_g, x^n_{-g}); k)\right] dF^{g,\hat{h}}(\hat{w})}_{g \text{ accepts new job}} + \underbrace{\mathbb{I}_{p_g=1}\tau}_{g \text{ exits parental leave}} \Big\} \Bigg].$$

Note that outflows due to voluntary quits to non-employment or entry into parental leave are already included in this expression, either as a consequence of a change of family state, or job acceptance of the partner, or the partner's return from parental leave.

Regarding the inflow into state (x_m, x_f, k), consider for example a household without young children ($k \neq 2$) where both members are non-employed ($h_m = h_f = 0$). The inflow into this state can involve skill depreciation, change of the family state with or without job

[6] Smoothing discrete choices by adding i.i.d. additive taste shocks is a standard device in dynamic discrete-choice models; Arcidiacono et al. (2022) employ such shocks in a continuous-time job-search model and show that they render the value functions analytically tractable. See also the numerical implementation in Fang and Shephard (2019, Appendix E).

quitting, layoffs, or the entry of new households into the first family state $k = 1$:

$$\mathcal{I}(x_m, x_f, k) \equiv \underbrace{\sum_{g=m,f} \pi^-_{x_g^+} G(x_g^+, x_{-g}, k)}_{\text{skill depreciation}} + \sum_{k_0} \Phi_{k_0 k} \Bigg\{ \underbrace{G(x_m, x_f, k_0)}_{\substack{\text{family state change} \\ \text{without job quitting}}}$$

$$+ \underbrace{\sum_{\hat{h}} \int G(\hat{x}_m, x_f, k_0)\Omega((x_m, x_f); (\hat{x}_m, x_f); k) + G(x_m, \hat{x}_f, k_0)\Omega((x_m, x_f); (x_m, \hat{x}_f); k) d\hat{w} \Bigg\}}_{\text{family state change where one partner quits job } (\hat{w}, \hat{h})}$$

$$+ \underbrace{\sum_{\hat{h}} \delta_{\hat{h}} \int G(\hat{x}_m, x_f, k) + G(x_m, \hat{x}_f, k) d\hat{w}}_{\text{one partner is laid off from job } (\hat{w}, \hat{h})} + \underbrace{\mathbb{I}_{k=1} N \zeta(s_m, s_f)}_{\text{entry of new couple households}} \quad .$$

In the last line, N is the inflow measure of couple households per unit time of which fraction $\zeta(s_m, s_f)$ enter the labor market with skills (s_m, s_f). The inflow expressions for the other couple households, as well as the inflow and outflow identities for single households are derived similarly. In a stationary equilibrium, inflows are equal to outflows, $\mathcal{I}(x_m, x_f, k) = \mathcal{O}(x_m, x_f, k)$ for all household states (x_m, x_f, k).

2.4 Firm Decisions

Firms choose how many jobs with hours requirements $h = h^F, h^P$ to create and the skill-adjusted wages they offer to workers of gender $g = m, f$ when encountering those workers. Firms anticipate that better paid jobs attract and retain more workers, as in Burdett and Mortensen (1998). In addition, they anticipate how wage policies change parental leave take-up and the option to switch from a full-time to a part-time job for parents of young children when parental leave ends. The firms' wage-offer policies take into account the gender wedges described above as well as the distinct labor supply behavior of the two genders (statistical discrimination).

If a firm with productivity y and a job with hours requirement $h \in \{h^F, h^P\}$ offers wage w_m to male workers and w_f to female workers, the perceived profit value of these job offers can be written

$$(y - w_m)\ell(m, h, w_m) + (y\eta^{F/P} - w_f)\ell(f, h, w_f) \ , \tag{2.2}$$

where $\ell(g, h, w_g)$, $g = m, f$, are the total efficiency units of labor of both genders that the firm expects to employ. To take parental leave (state $p = 1$) and the option of part-time

requests (state $p = 2$) into account, the actual working hours of a worker are

$$H_p(h) \equiv \begin{cases} h & \text{if } p = 0 \ , \\ 0 & \text{if } p = 1 \ , \\ h^P & \text{if } p = 2 \text{ and } h = h^F \ . \end{cases}$$

The efficiency units of labor that workers of gender g provide to a job offer (h, w_g) can be calculated by dividing total efficiency units of labor in those jobs by the total number of offers (h, w_g):

$$\ell(g, h, w_g) = \frac{\sum_{k,s,p} sH_p(h) \left[G^g(w_g, h, s, p, k) + \sum_{h_{-g}, s_{-g}, p_{-g}} \int G((w_g, h, s, p), (w_{-g}, h_{-g}, s_{-g}, p_{-g}), k) dw_{-g} \right]}{\Delta^h f^{h,g}(w_g)} \ , \tag{2.3}$$

with wage-offer density $f^{h,g}(w_g) \equiv dF^{g,h}(w_g)/dw$.

We break down the firm's decision into two steps. First, the firm chooses how many jobs with hours requirement h to create, and second, the firm chooses the skill-adjusted hourly wages w_g that it offers to both genders in these jobs.

Beginning with the second step, let $w_{g,h}(y)$ denote the optimal wage offer of firm y to a worker of gender g which maximizes the profit value in equation (2.2), so that the expected perceived profit of a job with hours requirement $h \in \{h^F, h^P\}$ is

$$\pi_h(y) \equiv (y - w_{m,h}(y))\ell(m, h, w_{m,h}(y)) + (y\eta^{F/P} - w_{f,h}(y))\ell(f, h, w_{f,h}(y)) \ . \tag{2.4}$$

Given these profit values, firm y decides which jobs h to offer. The job creation costs for a job with h required hours have two components: a common component $\bar{\varepsilon}_h$ and an idiosyncratic component ε_h which follows a type-I extreme value distribution with parameter σ_ε. A firm with productivity y chooses required hours to maximize

$$\max_h \{\pi_h(y) - \bar{\varepsilon}_h - \varepsilon_h\} \ ,$$

so that the proportion of productivity-y firms choosing $h = h_F, h_P$ is equal to

$$\Delta^h(y) = \frac{\exp[(\pi_h(y) - \bar{\varepsilon}_h)/\sigma_\varepsilon]}{\exp[(\pi_{h_F}(y) - \bar{\varepsilon}_{h_F})/\sigma_\varepsilon] + \exp[(\pi_{h_P}(y) - \bar{\varepsilon}_{h_P})/\sigma_\varepsilon]} \ . \tag{2.5}$$

The total fraction of job offers with $h = h_F, h_P$ is then calculated as

$$\Delta^h = \int \Delta^h(y)\ d\Gamma(y)\ ,$$

such that $\Delta^{h_F} + \Delta^{h_P} = 1$. We verify in numeric solutions of our model that wage offers $w_{g,h}(y)$ are monotonically increasing in firm productivity y. Therefore, the fraction of wage offers in jobs (g, h) below or equal to $w_{g,h}(y)$ is equal to the fraction of job offers (g, h) in firms with productivity y or less,

$$F^{g,h}(w_{g,h}(y)) = \frac{1}{\Delta^h} \int^{y} \Delta^h(y')\ d\Gamma(y')\ ,$$

which determines the equilibrium wage-offer distributions in jobs (g, h).

2.5 Stationary Equilibrium Definition

A stationary equilibrium is defined by value and policy functions for single and couple households, V^g, V, Ω^g, Ω, stationary distributions of single and couple households over labor market and family states, G^g and G, job creation strategies $\Delta^h(y)$ and wage offers $w_{g,h}(y)$ for all firms such that:

1. Households make optimal job acceptance, job quitting, parental leave and part-time request decisions.
2. Firms make optimal job offer decisions.
3. The household distributions are stationary.

We describe the numerical solution of the stationary equilibrium in Section C.1.

3 Calibration

3.1 Data

We calibrate our model using moments obtained from the German Socio-Economic Panel (GSOEP), a longitudinal survey that follows a representative sample of households in Germany throughout their life cycles. This dataset is particularly suitable for our study as it includes information on household composition and the labor market outcomes of adult household members. We use GSOEP data from 2007 to 2019, covering the period following

Table 3.1: External Parameters: Demographic Transition Rates

Transition	Couple	Single male	Single female	Target
$k = 1$ to $k = 2$	0.085	0.040	0.084	Age at first child
$k = 1$ to $k = 4$	0.010	0.046	0.044	Share never had children
$k = 2$ to $k = 3$	0.090	0.106	0.098	Age when youngest child turns 8
$k = 3$ to $k = 4$	0.123	0.095	0.086	Age when youngest child turns 20
Retirement rate	0.044	0.054	0.053	Retirement age 65

Note: This table reports demographic transition rates ($\Phi_{kk'}$) by household type. Targets are computed using GSOEP, 2007–2019.

the reform to parental leave and parental allowance policies. Appendix D provides further details on the sample, the variables used and the estimation approach. Additionally, we compute measures of search intensity across labor market and demographic states using the Panel Study Labour Market and Social Security (PASS) survey for the years 2007–2019.

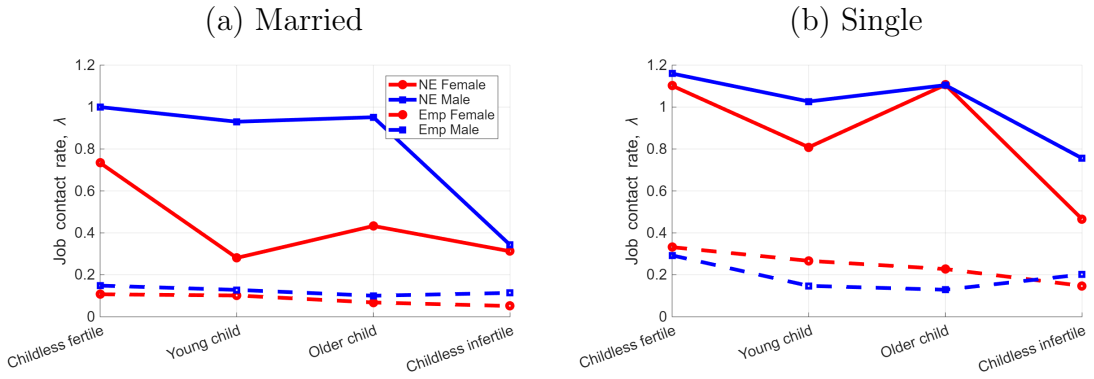
3.2 External Calibration

We set the annual discount rate ρ to 5%. Demographic transition rates are set exogenously as follows. Transitions from childless fertile to either a household with young children (Φ_{12}) or to childless infertile (Φ_{14}) are set to match the average age of the household head at the birth of the first child and the share of households that never have children. The transition rates Φ_{23} and Φ_{34} are set to match the average age of the household head when the youngest child turns 8 or 20, while Φ_{45} is set to match the retirement age 65. These rates are set separately for couple and single (male/female) households, while all other transition rates are zero. To ensure that the household distribution is stationary, the entry flow for each household type (N for couples and N^g for singles of gender g) equals the flow into the state of retirement ($k = 5$) of that household type. The resulting transition rates are reported in Table 3.1.

We assume that the set of required working hours H_+ includes two levels $\{h_{PT}, h_{FT}\}$, corresponding to part-time and full-time employment. We will use the time use data from GSOEP to calibrate these time endowment parameters. The hours are calibrated to the average working hours of each type of employment in GSOEP. Hours endowment $\bar{h}$ is assumed to be 80 hours per week. We are choosing two levels of skills where the lower skill level is set to reflect the high-to-medium skill premium in the German labor market (see e.g. Figure 1 in Glitz and Wissmann, 2021).

The parental leave exit rate τ is a policy parameter calibrated to the inverse of the median parental leave duration observed in GSOEP. We parametrize the tax-transfer functions Y^c

Figure 3.1: Job search effort by gender, demographic group, and marital status



Note: This figure shows the mean number of job applications, normalized by the value of married men in demographic state "childless, fertile" ($k = 1$), computed using PASS data for the years 2007–2019.

and Y^s with a log-linear specification (Heathcote et al., 2017) plus parental allowance. For singles, we assume that

$$Y^s(x_g, k) = \mathbb{I}_{\{p_g=0\}} \tau_0^s Y_g(x_g)^{1-\tau_1^s} + \mathbb{I}_{\{p_g=1\}} Y_p^s(x_g, k) \ , \tag{3.1}$$

where $Y_G(x_g) = w_g h_g s_g$ is the gross labor income, and $Y_p^s(x_g, k)$ is the parental allowance for a single parent with demographic type k. For couples, we assume that

$$Y^c(x_m, x_f, k) = \tau_0^c \left[\sum_{g=m,f} \mathbb{I}_{\{p_g=0\}} Y_G(x_g) \right]^{1-\tau_1^c} + \sum_{g=m,f} \mathbb{I}_{\{p_g=1\}} Y_p^c(x_g, k) \ , \tag{3.2}$$

reflecting joint income taxation at the household level. The parameters of the tax-transfer functions $\tau_0^s, \tau_1^s, \tau_0^c, \tau_1^c$ can be directly estimated from GSOEP by regressing log net income on log gross income, while excluding parental allowance (see Appendix D for further details).

Finally, we calibrate the offer arrival rates $\lambda_{g,k,\chi}^{e/n}$ ($g = m, f$, $k = 1, \ldots, 4$, $\chi = s, c$) of workers in different demographic and labor market states. We assume that job offers are proportional to search effort, which we measure by the mean number of job applications reported in the PASS survey. As shown in Figure 3.1, search effort differs little across the life cycle for most groups, but married mothers search markedly less than married fathers once children arrive. We set the offer arrival rates proportionally to these values, and list all further externally calibrated parameters in Table 3.2.

Table 3.2: External Parameters

Parameter	Description	Value	Target/Source
Preference parameters:			
γ	Labor supply elasticity	2	Standard value
$\bar{h}$	Time constraint (hours/week)	80	Standard value
ρ	Discount rate	0.05	Standard value
Skills			:
s_H	High skill	1	Normalization
s_L	Low skill	0.64	Glitz and Wissmann (2021)
Weekly hours:			
h_F	Weekly hours in FT	43.259	Mean weekly hours full time (GSOEP)
h_P	Weekly hours in PT	20.397	Mean weekly hours part time (GSOEP)
Policy parameters:			
b_m	NE benefit, Men	919.02	Median monthly unemployment benefit, Men (GSOEP)
b_f	NE benefit, Women	699.06	Median monthly unemployment benefit, Women (GSOEP)
$\tau_{0,k}$	Tax function, Level	55.72	Tax regression (GSOEP)
$\tau_{1,k}$	Tax function, Curvature	0.37	Tax regression (GSOEP)
	Parental leave expiration rate	0.5	1/median duration of parental leave
	Parental leave benefits	0.67	Replacement rate (GSOEP)
	Minimum PL allowance (per month)	100	Statutory policy
	Maximum parental leave (per month)	6000	Statutory policy

Note: This table reports the parameters externally computed using GSOEP, 2007–2019. Monetary values are expressed in 2015 euro.

3.3 Two-Step Calibration

The remaining parameters are calibrated through a two-step procedure following Bontemps et al. (2000) and Fang and Shephard (2019). In the first step, we parametrize the offer distributions and calibrate their parameters along with other supply-side parameters. We can compute labor supply given the calibrated offer distributions, which is used in the second step. In the second step, we recover the firm productivity distribution and the demand-side parameters.

3.3.1 Step 1: Calibration of Supply-Side Parameters

The supply-side parameter vector θ_S contains the utility parameters (ψ_k^g, α_k^g), the initial skill distribution ζ^g, skill-appreciation and skill-depreciation probabilities (π_x^+, π_x^-), separation rates δ_h with $\Delta^F = 1 - \Delta^P$, the parameters of the conditional wage-offer distributions $F^{g,h}(w)$, and the hours-offer distribution Δ^h.

For every candidate θ_S, we solve the single and couple household problems, construct the job-acceptance, quitting, parental-leave, return-to-work, and hours-reduction rules, and simulate household panels. The targeted moments include employment and hours distributions, wage quantiles, labor-market transitions, entry-wage quantiles, and wage growth. In what follows, we discuss how the targeted moments discipline the supply-side parameters.

For every gender $g \in \{m, f\}$ and hours requirement $h \in \{h^P, h^F\}$, the conditional

wage-offer distribution $F^{g,h}(w)$ is a Beta distribution on a calibrated support with shape parameters $a^{g,h}$ and $b^{g,h}$. The observed hourly wage distributions by gender and hours identify the wage-offer distribution parameters, while the observed part-time employment share identifies Δ^h.

The preference parameter α_k^g influences the employment level for gender g and demographic state k, while ψ_k^g influences the hours distribution given g and k. We normalize $\psi_{k=1}^{g=m} = 1$ and calibrate the remaining preference parameters to match part-time and full-time employment rates by gender and demographic state.

We assume that the skill grid has two points. The initial skill distribution $\{\zeta^g(1), \zeta^g(2)\}$, with $\zeta^g(1) + \zeta^g(2) = 1$, is identified from the gender-specific hourly wage distribution among full-time workers at age 25. Although ζ^g also influences the overall hourly wage distribution used to identify the wage-offer distribution parameters, the difference between hourly wages at age 25 and in the full sample identifies ζ^g separately from the wage-offer distribution parameters.

Skill appreciation π_x^+ is identified from year-to-year wage growth by hours type. Skill depreciation π_x^- is identified from the re-employment wage loss per month of non-employment. Separation rates δ_h are identified from transitions from each type of employment into non-employment.

Given the calibrated offer distribution, we can compute the labor-supply function $\ell(g, h, w)$ using equation (2.3). This function will be used in the next step of the calibration procedure.

3.3.2 Step 2: Calibration of Demand-Side Parameters

The demand-side parameter vector θ_D contains the gender wedges η^h, the hours-specific job-posting costs $\bar{\varepsilon}_h$, and the type-I extreme-value parameter σ_ε. We choose θ_D so that firms' optimal hours-posting decisions, $\Delta^h_{\text{th}}(y)$, reproduce the hours choices implied by the first-step offer distributions, $\Delta^h_{\text{ob}}(y)$. We refer to $\Delta^h_{\text{th}}(y)$ as the theoretical hours-posting decision and to $\Delta^h_{\text{ob}}(y)$ as the observed hours-posting decision.

The theoretical hours-posting decision $\Delta^h_{\text{th}}(y)$ can be computed using equation (2.5), which requires knowledge of the firm productivity associated with each posted wage and of the profit function $\pi_h(y)$. The first-order conditions for profit maximization with respect to wages yield the inverse wage equations, which allow us to recover firm productivity levels:

$$
\begin{aligned}
y = w^{-1}_{m,h}(w) &= w + \frac{\ell(m, h, w)}{\partial_w \ell(m, h, w)} \ , \\
y = w^{-1}_{f,h}(w) &= \frac{1}{\eta^h} \left[w + \frac{\ell(f, h, w)}{\partial_w \ell(f, h, w)} \right]
\end{aligned}
\tag{3.3}
$$

where labor supply $\ell(g, h, w)$ is computed from the first-step calibration. Inverting equation (3.3) yields the wage equation $w_{g,h}(y)$, which we use to compute the profit from posting hours h given productivity y as follows:

$$\pi_h(y) = [y - w_{m,h}(y)]\, \ell(m, h, w_{m,h}(y)) + \left[\eta^h y - w_{f,h}(y)\right] \ell(f, h, w_{f,h}(y)) \ . \tag{3.4}$$

We next construct the observed hours-posting decision $\Delta^h_{\text{ob}}(y)$. For each gender and hours type, define the conditional productivity distribution and density

$$\begin{aligned} Q^{g,h}(y) &= F^{g,h}(w_{g,h}(y)) \ , \\ q^{g,h}(y) &= f^{g,h}(w_{g,h}(y))\, w'_{g,h}(y) \ . \end{aligned} \tag{3.5}$$

In equilibrium, the conditional distribution of offers across productivity types is common across genders, so $Q^{m,h}(y) = Q^{f,h}(y)$ and $q^{m,h}(y) = q^{f,h}(y)$. Denote the common density for hours type h by $q^h(y)$. The joint density of productivity y and hours h implied by the first step is then $\Delta^h q^h(y)$. Summing across hours gives the firm-productivity density, and the first-step-implied hours-posting probability is the corresponding conditional share:

$$\gamma(y) = \sum_{\widetilde{h} \in \{h^P, h^F\}} \Delta^{\widetilde{h}} q^{\widetilde{h}}(y) \ , \qquad \Delta^h_{\text{ob}}(y) = \frac{\Delta^h q^h(y)}{\gamma(y)} \ . \tag{3.6}$$

We calibrate θ_D by minimizing the distance between $\Delta^h_{\text{th}}(y)$ and $\Delta^h_{\text{ob}}(y)$. To implement the two-step calibration, we discretize the support of wages and productivity. We relegate the implementation details of the calibration procedure to Section C.2. In particular, because the offer distribution recovered from the first-step calibration need not satisfy the requirement that the conditional productivity distribution of offers be common across genders, we discuss in the appendix how to approximate the common density function $q^h(y)$ given $q^{g,h}(y)$.

Once all model parameters are calibrated, we solve for the model equilibrium following the numerical procedure described in Section C.1 to ensure that the model moments are computed from the equilibrium distribution of the calibrated model.

3.4 Model Fit

Table 3.3 shows the internally calibrated parameter values and the calibration targets. Figure 3.2 shows that our model produces a good fit to the employment and hours distribution of male and female workers in the four demographic states. Figure 3.3 shows how our model fits the wage distribution of men and women in full-time and part-time jobs, while Figure 3.4

Table 3.3: Internal parameters

Parameter	Description	Target	Men	Women
Supply-side				
$\delta(P)$	Exog. sep. rate from PT	Monthly transition from PT to NE	0.080	
$\delta(F)$	Exog. sep. rate from FT	Monthly transition from FT to NE	0.006	
ψ_d^g: Preference for home hours				
	Childless, fertile ($d=1$)	Hours by (g,d)	1.000	2.953
	Young children ($d=2$)		0.008	10.509
	Older children ($d=3$)		0.008	8.888
	Childless, infertile ($d=4$)		0.223	2.614
α_d^g: Preference for employment				
	Childless, fertile ($d=1$)	NE rates by (g,d)	1.224	4.929
	Young children ($d=2$)		7.971	2.642
	Older children ($d=3$)		5.886	6.488
	Childless, infertile ($d=4$)		-0.215	1.597
$\zeta^g(2)/\zeta^g(1)$	Initial skill ratio (high/low)	Wage distribution of young, FT	0.415	0.593
$\pi^{P,+}$	Skill growth rate in PT	Annual wage growth, PT	0.278	
$\pi^{F,+}$	Skill growth rate in FT	Annual wage growth, FT	0.420	
π^-	Skill depreciation rate in NE	Re-employment wage loss per month of NE	0.076	
a^P	Beta shape, PT offer	Hourly wages, PT	1.312	
a^F	Beta shape, FT offer	Hourly wages, FT	1.604	
b^P	Beta shape, PT offer	Hourly wages, PT	3.583	
b^F	Beta shape, FT offer	Hourly wages, FT	3.583	
Δ^P	Part-time offer share	Share of PT jobs	0.219	
Demand-side				
$\bar{\epsilon}^P$	Fixed cost of posting PT job ($\bar{\epsilon}^F=0$)	Share of PT job offers	0.234	
σ_ϵ	Scale param. of job-posting cost	Share of PT offers by firm	0.210	
η^P	Gender wedge	Wages by gender, PT	–	1.273
η^F	Gender wedge	Wages by gender, FT	–	0.920

Note: This table reports the internally calibrated parameter values. † indicates that we are reporting the average across demographic types. NE, PT, and FT refer to non-employment, part-time, and full-time, respectively.

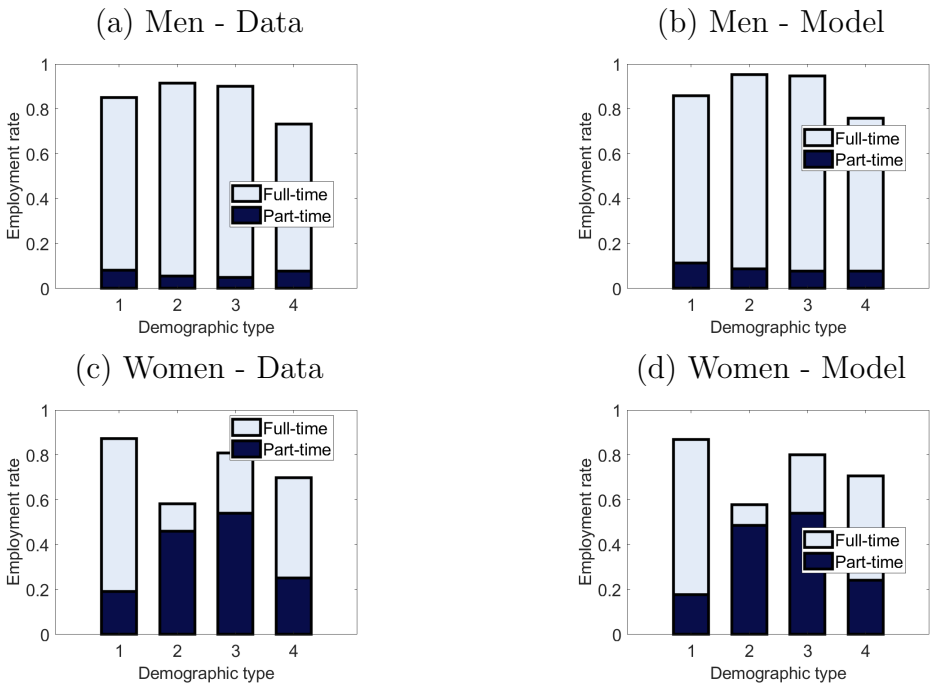
Table 3.4: Model fit: additional moments

	Data	Model
Job sep. prob, male PT	0.009	0.008
Job sep. prob, male FT	0.005	0.006
Wage growth, PT	0.0011	0.0009
Wage growth, FT	0.0017	0.0011

Note: Transition probabilities and wage growth are monthly. Wage growth is in log points.

shows the fit of the wage distribution of men and women at the time of labor market entry. Finally, Table 3.4 shows the model fit in terms of job separation rates and wage growth in full-time and part-time jobs. Section E.1 includes further model statistics: the distribution of workers by types of employment, including parental leave and reduced hours, and the male shares in part-time and full-time employment for all demographic groups.

Figure 3.2: Model fit: employment and hours distribution



Note: The height of the bars indicates the employment rate. Data moments are computed based on GSOEP 2007-2019. Demographic type 1 is childless fertile, type 2 has young children, type 3 has older children, and type 4 is childless infertile.

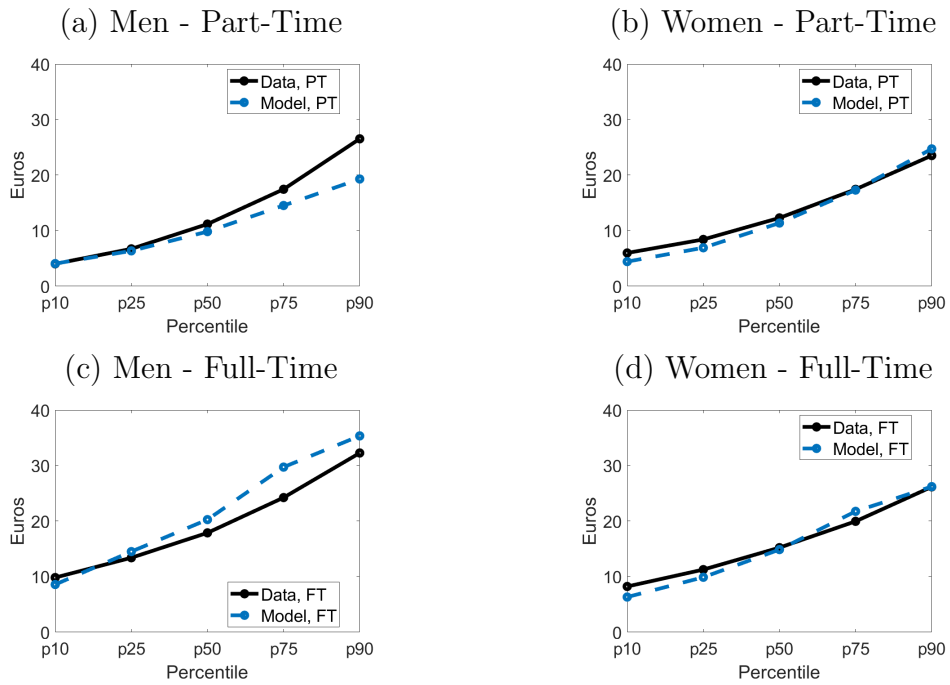
3.5 Model Validation

Gender Sorting. A key dimension that we do not target in the calibration is how men and women sort across firms of different productivity. In the model, firms choose which combinations of wages and working hours to offer, and workers sort across firms, so that firm heterogeneity directly shapes wage outcomes. In the model, productivity is the key dimension of firm heterogeneity which perfectly correlates with firm size. In the data and in the model, we group firms into three firm-size bins.[7] For each firm-size bin, we compute the share of workers by gender and parenthood status, distinguishing further between full-time and part-time employment. Figure 3.5 compares these workforce shares in the data (panels a and b) with those in the model (panels c and d). The model replicates the salient patterns in the data: women, and especially mothers, are disproportionately represented in smaller firms and are far more concentrated in part-time employment than men.[8] This

[7]GSOEP asks workers about the workforce size of their employing firm with seven response categories ($<$ 5, 5-10, 11-19, 20-99, 100-199, 200-1999, 2000+). We define the firm size bins in the model so that the employment share in each bin matches its counterpart in the data. For a more parsimonious representation, we use three firm size bins: 1-19, 20-199, and 200+ employees.

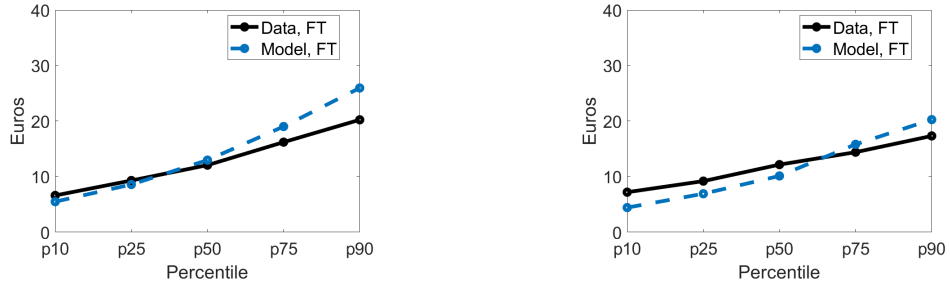
[8]The part-time concentration among mothers emerges sharply around childbirth. Using an event-study design, we estimate the transition into part-time employment around the birth of the first child and find

Figure 3.3: Model fit: hourly wage distribution by hours and gender



Note: Data moments are computed based on GSOEP 2007-2019.

Figure 3.4: Model fit: hourly wage distribution of young workers

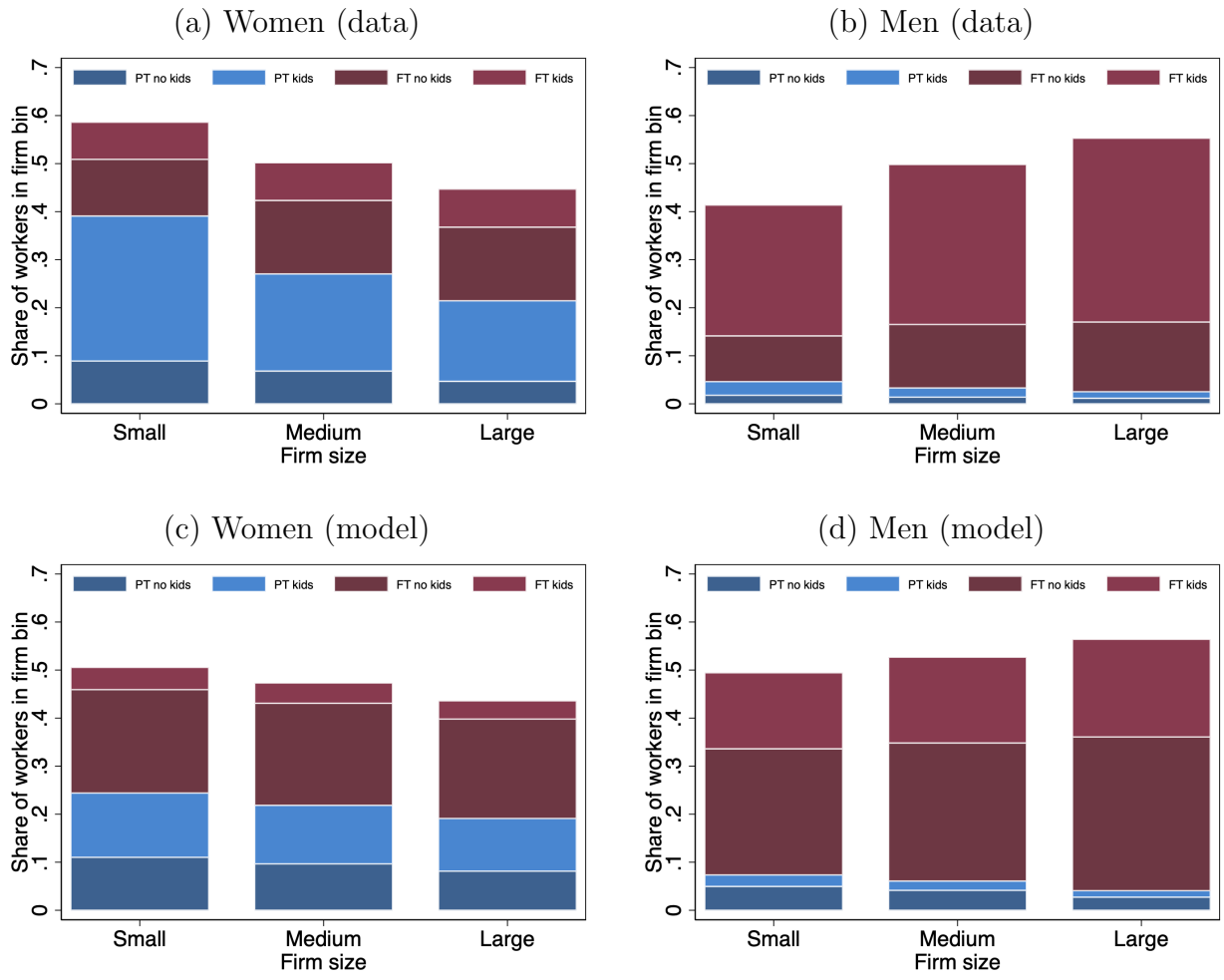


Note: Young workers are those age 25. Data moments are computed based on GSOEP 2007-2019.

sorting pattern reflects how firms design their jobs. Because the costs of creating a full-time or part-time job are the same for all firms while the output gain from full-time hours is larger in more productive firms, more productive firms post a greater share of full-time jobs (see Appendix Figure E.3). Since men, and especially fathers, work full-time far more often than women, they are more likely to be hired and retained by these high-productivity firms. As a consequence, women, and especially mothers, concentrate in less productive, lower-paying

that the probability of working part-time increases significantly for women but not for men; see Figure A.2.

Figure 3.5: Gender sorting across firms and working hours

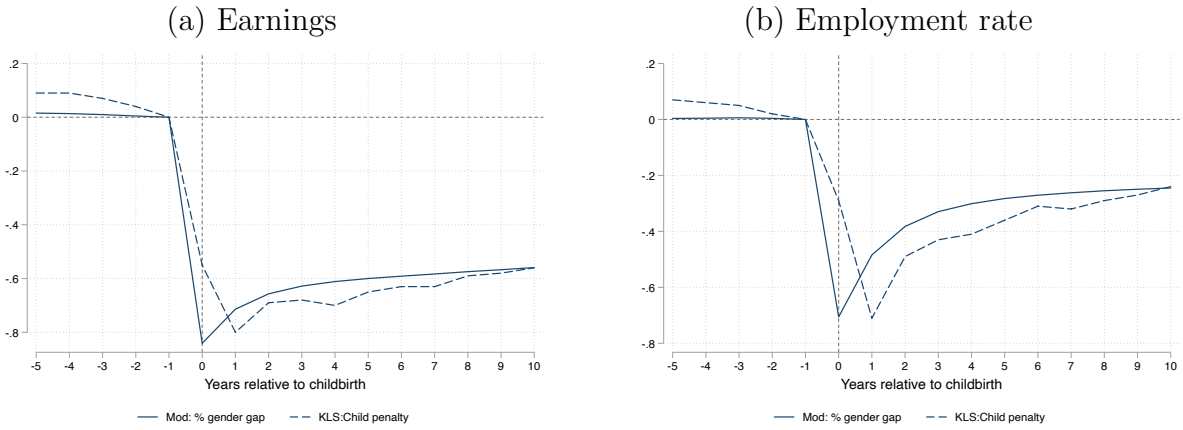


Note: The figure shows the share of workers in the aggregate workforce employed by firm-size bin. Small, medium, and large firms are defined as having below 20, between 20 and 199, and above 199 workers, respectively. Panels (a) and (c) report women; panels (b) and (d) report men. The top row (panels a and b) shows the data; the bottom row (panels c and d) shows the model. Blue and red bars indicate part-time and full-time jobs, respectively; darker and lighter shades represent workers without and with children, respectively. Data moments are computed from GSOEP, 2007–2019.

firms (see Appendix Figure E.5); this gendered sorting is strongest for parents with older children and persists even when all firms are constrained to post the same share of part-time jobs (see Appendix Figure E.4). Because both the firm-size wage premium and the full-time/part-time hourly wage gap are larger for men than for women (see Figure A.1), the overrepresentation of women, and especially mothers, in part-time positions and in smaller firms contributes to the gender wage gap, as we show in Section 4.2.

Gender Gaps around Childbirth. Next, we examine whether the model is able to reproduce the evolution of the gender gap in labor market outcomes around the birth of the first child documented by Kleven et al. (2019a). To facilitate a direct comparison with

Figure 3.6: Gender gap around childbirth



Note: The figure compares the gender gap in earnings (panel a) and employment rate (panel b) around the birth of the first child in the model and in the data. For the model, the gender gap is constructed from a simulated panel as the difference between the estimated average of the reference variable for women and men normalized to the year before childbirth. For the data, the gender gap is computed as the difference between the estimated child penalties for women and men reported by Kleven et al. (2019a) and computed using the event-study methodology of Kleven et al. (2019b) ("KLS") on GSOEP data, with event time normalized to the year before the first birth ($t = -1$).

the empirical evidence, we apply the same event-study approach to the simulated panel generated by the model. Specifically, we identify the first transition to parenthood for each household, construct annual event time relative to the birth of the first child, and compute the average earnings and employment after normalizing each series to the year immediately preceding childbirth ($t = -1$). We then calculate the gender gap as the difference between the estimated child penalties for women and men at each event time, mirroring the procedure in Kleven et al. (2019b). Figure 3.6 compares the resulting gender gaps from the model with the empirical estimates for Germany reported by Kleven et al. (2019a), which apply the methodology of Kleven et al. (2019b) to the GSOEP.[9] The model reproduces the main features of the empirical evidence well. In both the data and the model, the gender gap in earnings widens sharply after childbirth, and remains persistently large over the subsequent years. A similar pattern emerges for employment, with a sizeable and long-lasting decline in women's employment relative to men's following parenthood. Quantitatively, the model closely matches the long-run gender gap in earnings, generating a penalty of 58% compared with 61% in the data. Likewise, the model produces a long-run employment penalty of 26%, compared with about 30% in the data. One discrepancy concerns the timing of the adjustment: the highest gender gap manifests on impact in the model and at time one in

[9]We additionally estimated the event-study profiles directly using our GSOEP sample following the methodology of Kleven et al. (2019b). The resulting estimates are quantitatively very close to those reported by Kleven et al. (2019a).

Table 4.1: Decomposition counterfactuals

	Worker	Firm	
Counterfactual	Search, init. skill, pref.	Gender wedge	Offer dist.
Benchmark	•	•	Eqm
Gender wedge	=	•	Eqm
Supply + statistical discr.	•	=	Eqm
Pure supply	•	=	=

Note: "=" indicates that the corresponding parameters are equalized between men and women (set to their average value), "•" indicates the actual gender-specific benchmark values, and "Eqm" indicates that the wage-offer distribution is determined in equilibrium. In the "Pure supply" counterfactual the wage-offer distribution is instead fixed at the one that prevails when all gender differences are eliminated.

the empirical estimates. This difference is expected because the simulated event-study is constructed from annual observations of the model, whereas fertility events and labor supply decisions occur continuously. Annual sampling therefore introduces a coarser measure of event time, implying that changes occurring within a year are recorded at the nearest annual observation. Despite this minor timing mismatch, the model captures both the magnitude and persistence of the gender gaps observed in the data, providing strong evidence that it successfully replicates the key labor market consequences of parenthood. Figure E.9 reports the underlying estimates of child penalties separately for men and women.

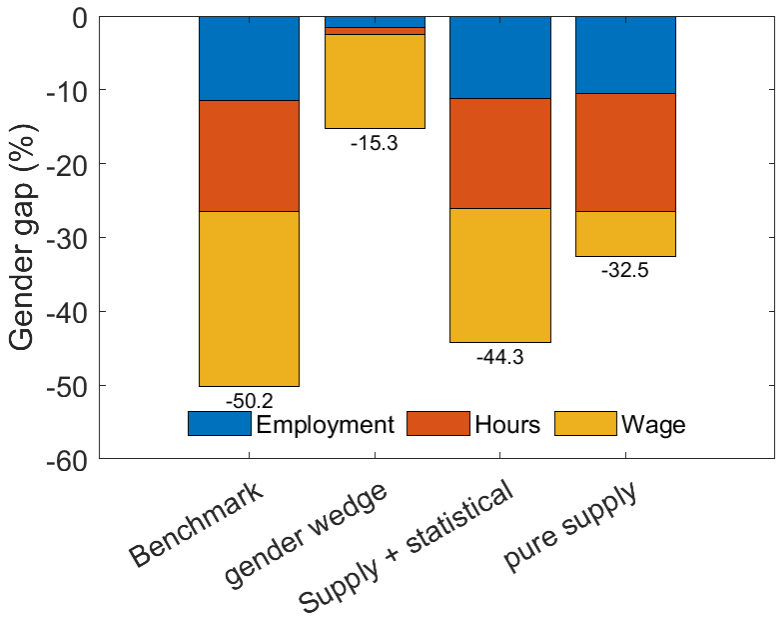
4 Results

4.1 The Gender Earnings Gap

To understand the mechanisms underlying gender inequality in earnings, we conduct a series of counterfactual exercises in which we eliminate specific gender differences and recompute the equilibrium. Gender differences emerge in our model through supply and demand channels: worker-side differences in job search, initial skills and preferences, and firm-side gender wedges and statistical discrimination. Table 4.1 summarizes the counterfactuals. In each of them we either equalize a channel between men and women (indicated by "=") or keep it at its benchmark, gender-specific value (indicated by "•"), while firms may adjust their wage offers in equilibrium ("Eqm").

The "Gender wedge" counterfactual equalizes all worker-side parameters, so that the remaining gender differences are driven by the firms' gender wedges. Conversely, the "Supply + statistical discrimination" counterfactual equalizes the gender wedges, so that gender differences arise from labor supply (job search, initial skills and preferences), to which

Figure 4.1: Gender gap decomposition



Note: Each bar shows the gender earnings gap, i.e. the percentage difference between mean female and male earnings, in the benchmark economy and in the three counterfactuals of Table 4.1. The gap is decomposed into the contributions of gender differences in employment, hours (conditional on employment) and hourly wages.

firms respond with gender-specific wage offers; this scenario therefore features statistical discrimination. To isolate the separate role of statistical discrimination, the "Pure supply" counterfactual additionally fixes the wage-offer distribution at the one that prevails when all gender differences are eliminated, so that men and women earn the same wages. Hence, gender gaps in the "Pure supply" counterfactual emanate exclusively from the worker-side parameters. The difference between the "Supply + statistical discrimination" and "Pure supply" scenarios thus measures the contribution of statistical discrimination.

Figure 4.1 compares the gender earnings gap in the benchmark economy with the three counterfactuals, decomposing each gap into the contributions of employment, hours (conditional on employment) and hourly wages. In the benchmark economy, female earnings are about 50 percent below male earnings, reflecting gender gaps in employment (about 23 percent), in hours conditional on employment (about 30 percent) and in hourly wages (about 47 percent). Comparing the counterfactuals, labor supply differences account for the bulk of the earnings gap (about 60 percent), while the firms' gender wedges and statistical discrimination account for about 21 and 19 percent, respectively. We obtain these contributions from a Shapley-Owen-Shorrocks decomposition of the earnings gap (Audoly et al., 2025), which we describe in Section E.3.

The "Gender wedge" scenario isolates the demand side: equalizing all worker-side differences roughly halves the wage gap and substantially compresses the employment and hours gaps. Gender wedges thus generate a sizable wage gap which induces a modest labor supply response at the extensive and intensive margin. Conversely, the "Supply + statistical discrimination" scenario reproduces almost the entire employment and hours gaps, confirming that these margins are supply-driven. The comparison with the "Pure supply" scenario isolates statistical discrimination: because firms' wage offers respond to gender differences in labor supply, this channel alone accounts for over ten percentage points of the hourly wage gap.

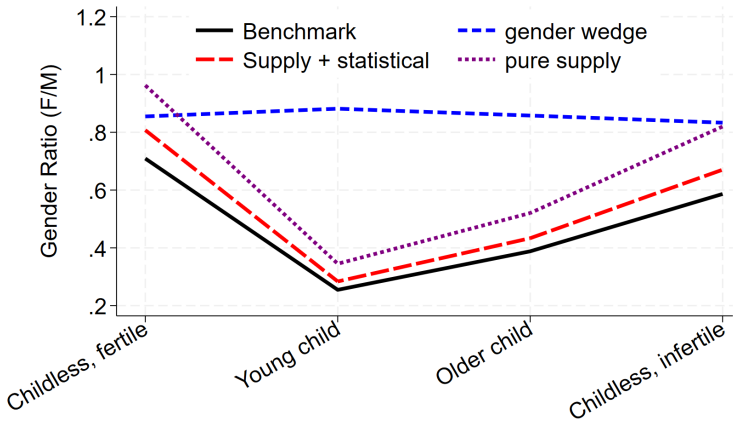
Decomposing the supply side further, Figure E.2 in the Appendix considers each supply factor in isolation. Gender differences in preferences are by far the most important, generating an earnings gap of about 43 percent on their own, whereas differences in job search contribute about 9 percent and the impact of differences in initial skills is negligible.

The gender earnings gap also varies systematically over the life cycle. Figure 4.2 shows the female-to-male earnings ratio across the four demographic states in the benchmark economy and the three counterfactuals. In the "Gender wedge" scenario the earnings ratio is essentially flat: the demand-driven gap is roughly constant and does not respond to the arrival of children. In contrast, the "Supply + statistical discrimination" and "Pure supply" scenarios both display a pronounced drop in the earnings ratio when a young child arrives, followed by a gradual recovery. Hence the widening of the earnings gap around childbirth is driven by supply factors, while demand-side forces (including statistical discrimination) generate a constant gap. Appendix Figure E.1 shows the corresponding decomposition into employment, hours and hourly wages: the worker side (gender differences in job search and preferences) accounts for the larger gaps in employment and hours in households with young and older children, whereas the wage gap arising from gender wedges is roughly constant across demographic states.

4.2 Gender Gap in Hourly Wages

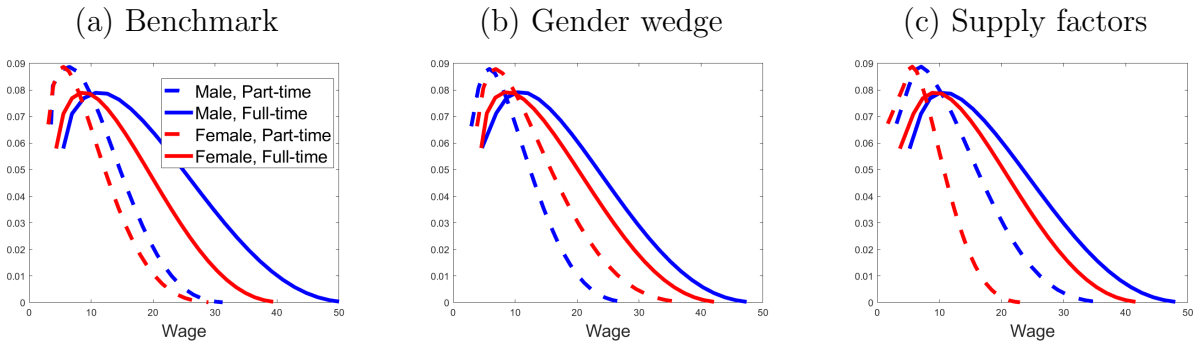
We now examine the gender gap in hourly wages in more detail. Figure 4.3 shows the distributions of firms' skill-adjusted wage offers to men and women in full-time and part-time jobs, and separately for the benchmark calibration and the demand- and supply-side counterfactuals. In our calibrated model, the wage gap in full-time jobs is particularly pronounced in the middle and at the top of the distribution, to which the firm side and the worker side contribute in roughly similar magnitudes. The wage gap in part-time jobs is much smaller. Isolating the firms' gender wedges in panel (b) reveals that when only gender wedges

Figure 4.2: Earnings gap over the life cycle



Note: Female-to-male earnings ratio (in %) across the four demographic states, in the benchmark economy and the three counterfactuals of Table 4.1.

Figure 4.3: Wage-offer distributions

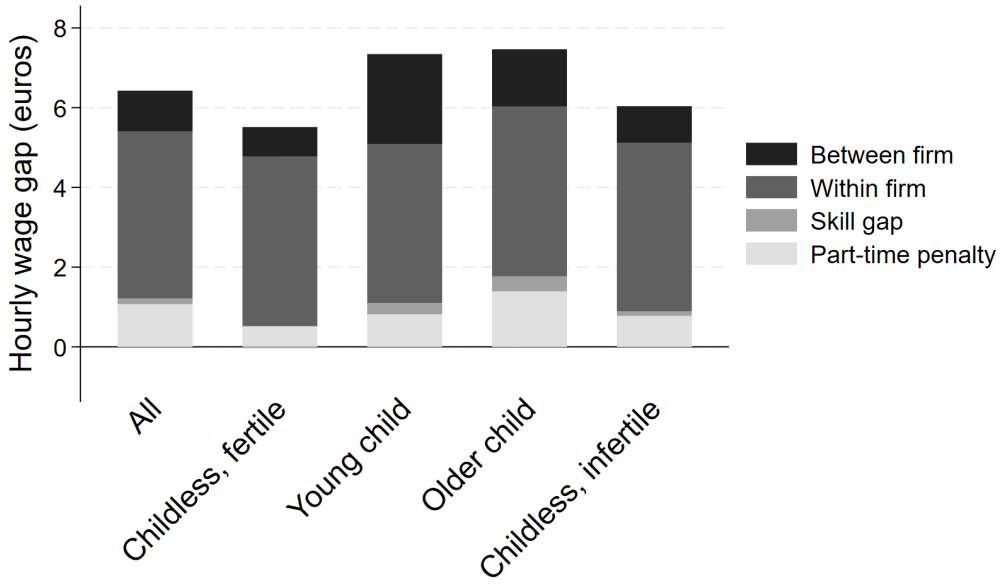


Note: Distribution of skill-adjusted wage offers to men and women in full-time and part-time jobs. Panel (a) shows the benchmark economy; panel (b) the "Gender wedge" counterfactual; panel (c) the "Supply + statistical discrimination" counterfactual.

are active, men earn *lower* wages than women in part-time jobs, i.e. firms attach a part-time wage penalty to male workers. Panel (c) isolates the supply side: when gender differences in preferences, job search and initial skills are taken into account while gender wedges are absent, a substantial wage discount for female workers arises. The explanation is the low female labor supply elasticity in part-time jobs which results in large wage markdowns.

To quantify the sources of the hourly wage gap, we decompose the mean gender gap into four components. Let $\bar{w}_g$ denote mean hourly wages of gender g, and let $\bar{w}_f^h$, $\bar{w}_f^{h,s}$ and $\bar{w}_f^{h,s,w}$ denote counterfactual mean female wages obtained by giving women men's working hours; men's working hours and skills; and men's working hours, skills and within-firm wages,

Figure 4.4: Decomposition of the gender gap in hourly wages over the life cycle



Note: Decomposition of the mean gender gap in hourly wages, following equation (4.1).

respectively. Then the mean gender gap in hourly wages is decomposed into

$$\bar{w}_m - \bar{w}_f = \underbrace{\bar{w}_m - \bar{w}_f^{h,s,w}}_{\text{between firm}} + \underbrace{\bar{w}_f^{h,s,w} - \bar{w}_f^{h,s}}_{\text{within firm}} + \underbrace{\bar{w}_f^{h,s} - \bar{w}_f^{h}}_{\text{skill gap}} + \underbrace{\bar{w}_f^{h} - \bar{w}_f}_{\text{part-time penalty}} \quad . \tag{4.1}$$

The first term is the gap due to men and women working in different firms (between-firm gender sorting), the second the gap in wages paid within the same firms, the third the gap due to differences in skills, and the fourth the penalty from women's higher incidence of part-time work. The within-firm component is by far the largest, accounting for about 65 percent of the gap, followed by the part-time penalty (about 17 percent) and the between-firm component (about 16 percent), while the skill gap contributes only about 2 percent.

Figure 4.4 shows how this decomposition varies over the life cycle. After childbirth the wage gap widens, driven mainly by the between-firm component, as women increasingly sort into lower-paying firms. The part-time penalty additionally contributes to the rising wage gap, particularly for women with older children. The skill gap also widens after childbirth, though it remains quantitatively less important.

4.3 The Role of Household Specialization

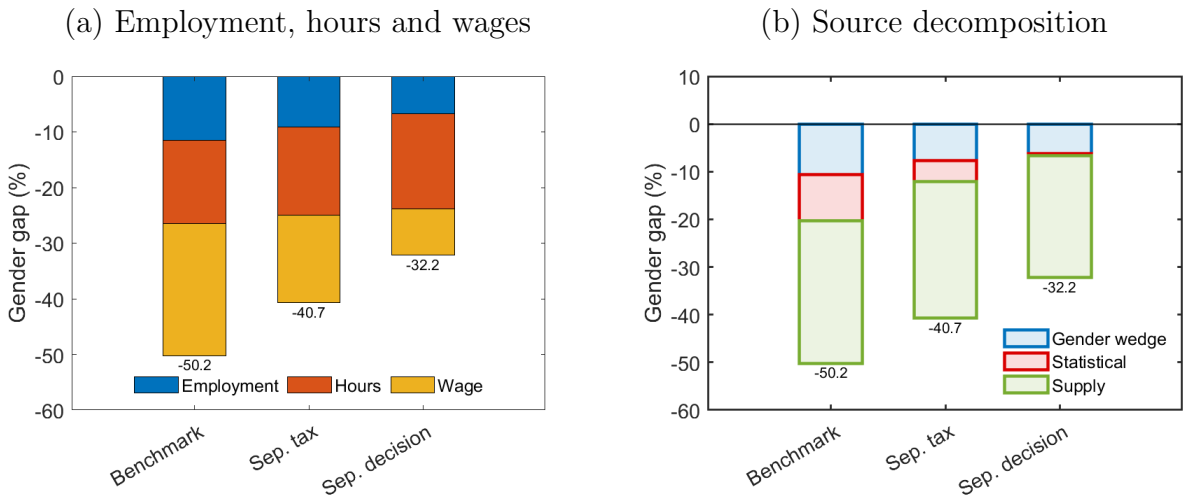
In this subsection, we aim to understand how the joint decision making in households affects the gender earnings gap. Our model has income pooling within the household which arises because of two features. First, income is jointly taxed at the level of the household, i.e. the sum of both spouses' earnings is taxed with a progressive tax schedule at the household level. Second, after-tax earnings are equally shared between the two spouses who then consume equal amounts, which is a consequence of intra-household risk sharing. Both features have implications for the labor supply of workers in couple households. First, as already pointed out by Pilossoph and Wee (2021), reservation wages of both spouses are higher when household income is pooled. We show that this feature makes labor supply of couple workers more elastic, in comparison to single workers who do not pool their income with a spouse. In Section B.4, we establish this finding using simple static labor supply models at the intensive and extensive margins. Second, there are differential responses for primary and secondary earners. With joint taxation and risk sharing (income pooling), an extra euro paid to either spouse has the same effect on household utility, regardless of who earns that extra euro. If instead the two spouses pay progressive taxes separately and if they do not share after-tax income equally, then an extra euro paid to the higher-income spouse has a smaller effect on household utility in comparison to an extra euro paid to the lower-income spouse. As men earn higher wages than women, the absence of income pooling should imply a larger drop of the reservation wages of men than the reservation wages of women, due to the smaller marginal effect of the male wage on household utility.

To quantify these forces and the response of firms to the altered labor supply, we consider two counterfactual economies that progressively remove household income pooling. In the *separate taxation* scenario, couples continue to share income risks, but each spouse's labor income is taxed separately with its own tax schedule. In the *separate decision* scenario, in addition to separate taxation, the spouses no longer share income risks; as a consequence, job acceptance and job quitting decisions are made separately by each spouse rather than jointly by the household. The separate decision scenario therefore removes intra-household insurance entirely, whereas separate taxation removes only its tax-side component.

We implement the separate tax schedules using the same two-parameter tax function and the same progressivity parameter as the joint tax schedule, shifting only the level parameter. Specifically, we set the level parameter of the separate schedule such that a couple household with median household income and median male earnings pays the same total taxes as under joint taxation. Section B.3 describes this procedure in detail.

Figure 4.5 summarizes the results. Panel (a) decomposes the gender earnings gap in

Figure 4.5: The role of income pooling for the gender earnings gap

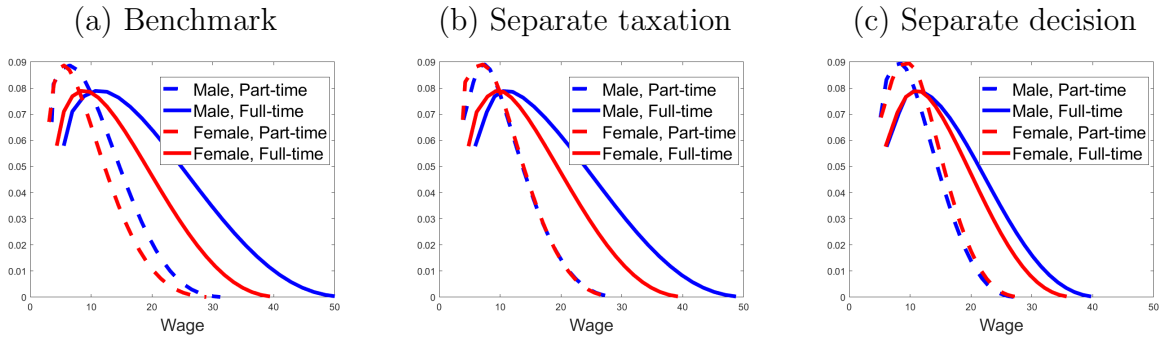


Note: Panel (a) decomposes the gender earnings gap (in %) in the benchmark (joint-decision) economy and the two counterfactuals into gender differences in employment, hours and hourly wages. Panel (b) decomposes the earnings gap in each economy into the contributions of the firms' gender wedge, statistical discrimination and pure supply, using the Shapley-Owen-Shorrocks decomposition described in Section E.3 (with the underlying numbers in Appendix Table E.3).

the benchmark (joint-decision) economy and in the two counterfactuals into its employment, hours and wage components. The gender earnings gap declines substantially under separate taxation and even more so under separate decisions, falling from about 50 percent in the benchmark to 41 percent under separate taxation and 32 percent under separate decisions. Some of this decline reflects smaller gaps in employment and hours. Figure E.6 reports the employment and hours distributions by gender and demographic state across the three economies. Women, and especially mothers, work more when household income is no longer pooled. The expansion in female employment is strongest in part-time jobs, while childless women also work more full time. In contrast, men expand their labor supply only slightly.

The largest contributor to the declining earnings gap in panel (a) is the wage component. The reason can be understood from the firms' wage offers. Figure 4.6 shows the equilibrium distribution of skill-adjusted wage offers to men and women in full-time and part-time jobs, in the benchmark economy (panel a) and in the two counterfactuals (panels b and c). Firms offer lower wages, especially in the separate-decision counterfactual, with the steepest wage-offer decreases to men in full-time employment. The mechanism operates through shifted labor supply. As explained above, removing income pooling lowers reservation wages, and it does so more for men than for women. Male labor supply therefore becomes less elastic, particularly in full-time jobs; Appendix Figure E.7 and Figure E.8 display the labor supply schedules and their elasticities for men and women. Because firms set wages as a markdown on productivity that is larger when labor supply is less elastic, wage markdowns rise, especially

Figure 4.6: Firms offer lower wages, especially to men

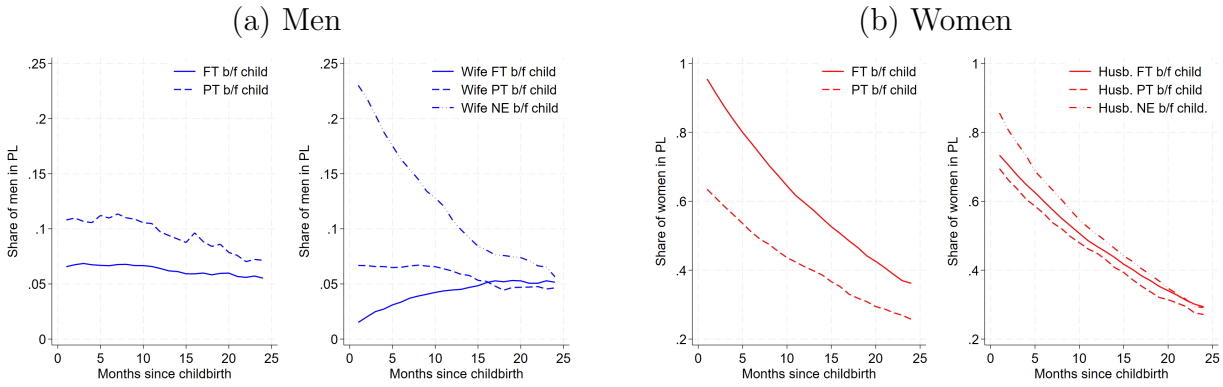


Note: Equilibrium distribution of skill-adjusted wage offers to men and women in full-time and part-time jobs. Panel (a) shows the benchmark (joint-decision) economy, panel (b) the separate-taxation economy, and panel (c) the separate-decision economy.

for men in full-time jobs. The effect is strongest in the separate decision scenario, where spousal insurance is absent.

Panel (b) of Figure 4.5 decomposes the earnings gap in each economy into the source contributions of the firms' gender wedges, statistical discrimination and pure supply. We compute these decompositions as in Section 4.1 and described in Section E.3, with the underlying numbers reported in Appendix Table E.3. Pure supply effects contribute modestly to the falling earnings gap: relative to the benchmark, the pure-supply component shrinks by 4.4 percentage points under separate decisions and by only 1.4 percentage points under separate taxation. The bulk of the decline instead comes from statistical discrimination, whose contribution to the earnings gap collapses from about ten percentage points in the benchmark to essentially zero under separate decisions. In other words, statistical discrimination arises almost entirely in response to household income pooling which makes especially male labor supply less elastic, leading firms to apply smaller wage markdowns, particularly to men in full-time jobs. Finally, the contribution of the gender-wedge term is larger under joint decision making than in the counterfactuals. This is because the gender-wedge term also embeds an endogenous response of workers to these wedges and therefore an additional statistical-discrimination component: firms take the worker responses into account when deciding their wage-offers. Our results show that statistical discrimination amplifies the effect of the gender wedges when households are subject to joint taxation and share income risks.

Figure 4.7: Parental leave take-up by own hours and spousal employment state



Note: Share of married fathers (panel a) and married mothers (panel b) on parental leave in the benchmark model, by months since childbirth. In each panel, the left chart reports take-up by the worker's own working hours prior to childbirth (full-time or part-time), and the right chart by the spousal employment state prior to childbirth.

4.4 Parental Leave Policy

4.4.1 Parental Leave: Take Up and Follow Up

In our benchmark model, 76 percent of mothers with young children take up parental leave, while this share of fathers is only 16 percent. Note that both these shares include a potential follow-up parental leave spell in a couple when the parental leave spell of their spouse ends. The average duration of parental leave in our model is 13.5 months for married mothers and only 0.4 months for married fathers.

Parental leave take-up in our model depends both on a worker's own working hours prior to childbirth and on the labor market state of the spouse. Figure 4.7 illustrates these patterns in the benchmark model for men (panel a) and women (panel b). The left charts in both panels report take-up by own working hours prior to childbirth: fathers are less likely to take up parental leave from a previous full-time job, whereas the opposite holds for mothers. As a result, married mothers are particularly more likely to leave their previous full-time job after childbirth than their full-time employed husbands. The right charts in both panels show that parental leave take-up also varies with the spousal employment state.

While the institutional framework allows parents to return to their previous job (at the same or reduced working hours) upon parental leave termination, the ultimate decision whether or not to return to the previous job or whether to request reduced hours varies across genders and also depends on the spousal labor market state. Table 4.2 shows the follow-up labor market status after parental leave termination for mothers (panel a) and fathers (panel b). Each row is a probability distribution of the subsequent employment

Table 4.2: Decision upon parental leave termination by spousal employment state

(a) Female PL termination

	FT	PT	PT(reduced)	NE
NE spouse	0.063	0.160	0.746	0.031
PT spouse	0.007	0.131	0.754	0.107
FT spouse	0.004	0.148	0.742	0.106

(b) Male PL termination

	FT	PT	PT(reduced)	NE
NE spouse	0.628	0.147	0.198	0.027
PT spouse	0.664	0.143	0.126	0.068
FT spouse	0.174	0.072	0.145	0.609

Note: Based on simulated data. Each row represents the probability distribution across four possible decisions when one's PL terminates. NE, PT, and FT refer to non-employment, part-time, and full-time, respectively. PT (reduced) refers to part-time employment as a result of hours reduction from a full-time job.

choice (return to full-time job, return to part-time job, request part-time from previous full-time job, or quit to non-employment), and it is reported separately by the spouse's labor market status (non-employment, part-time and full-time employment). It can be seen that mothers return much less frequently to previous full-time jobs and request much more frequently part-time employment compared to fathers. Mothers even do so when their spouse is non-employed. Fathers, in contrast, often decide to quit their job in those cases where their spouse is full-time employed.

4.4.2 Policy Counterfactuals

Current parental leave regulation in Germany stipulates the right to return to the previous job (possibly at reduced hours) and it pays a transfer (parental allowance) which is linked to the previous wage subject to a cap. To evaluate the separate effects of both features, we consider the following economies with counterfactual policies:

- Low replacement rate: the parental leave allowance replacement rate is half of the benchmark level, while the cap remains the same.
- Short parental leave: the parental-leave expiration rate (parameter φ) is twice as high as in the benchmark economy.

We also consider a combined scenario, and in all counterfactuals we calculate the stationary equilibrium of the economy in general equilibrium.

Table 4.3: Parental leave take-up and duration under policy counterfactuals

	Benchmark	Lower Replacement	Short PL	Short PL + Lower Repl.
Share parental-leave take-up (in percent)				
Married fathers	16.0	13.5	40.8	36.8
Married mothers	76.4	75.7	77.3	76.6
Average parental leave duration (months)				
Married fathers	0.4	0.3	1.9	1.5
Married mothers	13.5	13.5	7.5	7.4

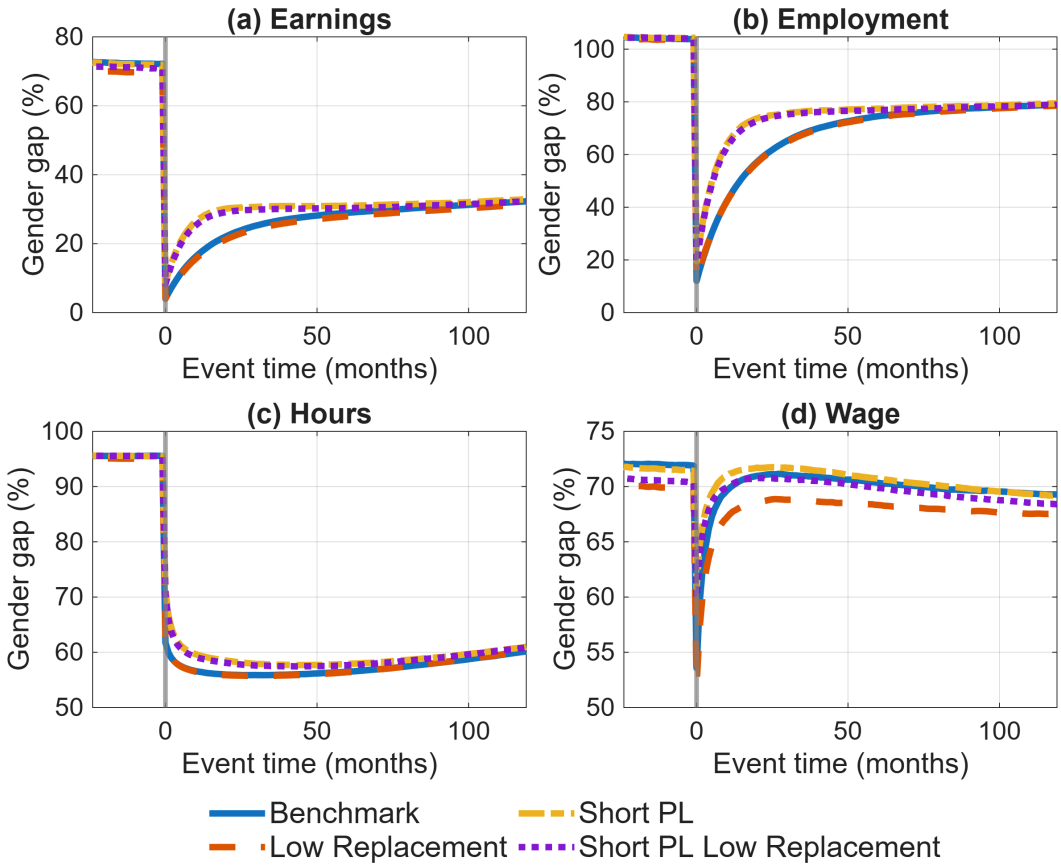
Note: Based on simulated data from the benchmark model and the three policy counterfactuals. "Lower Replacement" halves the parental-leave replacement rate, "Short PL" doubles the parental-leave expiration rate, and "Short PL + Lower Repl." combines both. Average parental leave duration is measured unconditionally on take-up, i.e. it averages over all married mothers or fathers.

Table 4.3 reports parental leave take-up and average duration in the benchmark and the three policy counterfactuals. When the replacement rate of parental leave is cut by half, the responses in take-up are small: the fathers' take-up rate falls from 16 to about 14 percent, while the mothers' take-up rate is essentially unchanged. The relatively larger decline of fathers can be explained by the observation that they earn higher wages prior to childbirth, so that the opportunity costs of parental leave are larger under the lower replacement rate. By contrast, when the parental leave duration becomes half as large, the share of fathers on parental leave increases substantially, from 16 to 41 percent, while the share of mothers remains high and virtually unchanged. The intuition is that many more men find it attractive to choose a short parental leave spell when this option becomes available. These take-up responses are mirrored in the average duration of parental leave: shortening the leave period roughly halves the mothers' average leave, from 13.5 to 7.5 months, while it raises the fathers' average leave from 0.4 to 1.9 months; lowering the replacement rate, by contrast, leaves average durations essentially unchanged. In the combined scenario (last column), lowering the replacement rate slightly dampens the strong take-up response of fathers to shorter leave.

Figure 4.8 shows how the gender gaps in earnings, employment, hours and hourly wages evolve in the ten years after childbirth, comparing the benchmark model with the policy counterfactuals. In line with the rather small changes in take-up rates, lowering the replacement rate has only a very small effect on these gaps, as few fathers and mothers change their parental-leave decision.

Shortening the duration of parental leave has a stronger (and negative) effect on the earnings of fathers who now choose much more frequently to take up parental leave. At the same time, mothers return to employment faster. Taken together, gender gaps in earnings,

Figure 4.8: Gender gaps before and after childbirth: policy counterfactuals



Note: Gender gap (married female to married male ratio, in %) in earnings, employment, hours (conditional on employment) and hourly wages in the ten years around childbirth, comparing the benchmark model with the policy counterfactuals.

employment and hours are somewhat smaller. As a result of higher employment and working hours, human capital losses of mothers are lower which is reflected in a faster recovery of hourly wages. On the other hand, the human capital losses of fathers are relatively small, as they leave the job only for a relatively short time. Indeed, we confirm that their hourly wages are nearly the same as in the benchmark economy.

5 Conclusions

This paper develops an equilibrium household-search model of the labor market that explicitly incorporates firms' endogenous decisions over job offers, including wage and hours requirements, in response to workers' life-cycle labor supply patterns. By allowing the distribution of part-time and full-time job offers to emerge from the interaction between workers'

evolving preferences and constraints and firms' strategic posting decisions, our framework provides new insights into the mechanisms underlying child-related gender gaps in labor market outcomes.

Our decomposition attributes about 60 percent of the gender earnings gap to labor supply factors and the remaining 40 percent to labor demand, split between the firms' gender wedges (about 21 percent) and statistical discrimination (about 19 percent). Worker-side factors, and especially gender differences in preferences for home hours, account almost entirely for the gaps in employment and working hours, while both gender wedges and statistical discrimination shape the hourly wage gap. These forces play out very differently over the life cycle: the demand-side contribution is roughly constant, whereas the supply side drives the pronounced widening of gender gaps around childbirth.

Two further findings stand out. The first concerns the sources of the hourly wage gap and how they evolve over the life cycle. On average, the gap is largely a within-firm phenomenon, but around childbirth its between-firm component increases, as mothers sort into lower-paying firms, while the part-time wage penalty adds further to the widening gap. The second concerns the origins of statistical discrimination. Because spouses pool income through joint taxation and intra-household risk sharing, household specialization makes labor supply of men relatively elastic, so that firms apply smaller wage markdowns, particularly to men in full-time jobs. When we remove income pooling in counterfactuals, the contribution of statistical discrimination to the earnings gap essentially disappears.

Our analysis of parental leave policies yields more tempered conclusions. Halving the leave duration has large behavioral effects, raising fathers' leave take-up substantially. Nevertheless, the effects on the gender gaps in earnings, employment, hours and wages are modest and short-lived. These results suggest that reforms to the design of parental leave, on their own, have limited scope to close the gender gaps that open up around childbirth.

More broadly, our results suggest that understanding gender gaps requires careful attention to the equilibrium structure of the labor market. Policies that affect workers' labor supply decisions will induce responses in firms' job posting strategies, which in turn feed back into workers' opportunities and choices. Future research might extend our framework to consider additional dimensions of job flexibility and job amenities, the role of workplace culture and norms, and the potential for policies that directly target firms' employment practices. As labor markets continue to evolve and policy debates around work-family balance intensify, equilibrium models that take seriously both sides of the market will remain essential tools for rigorous policy evaluation.

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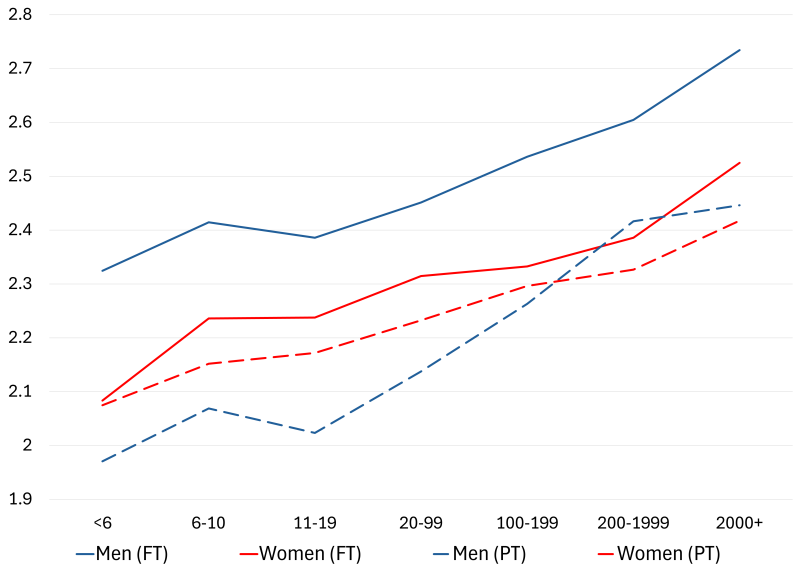
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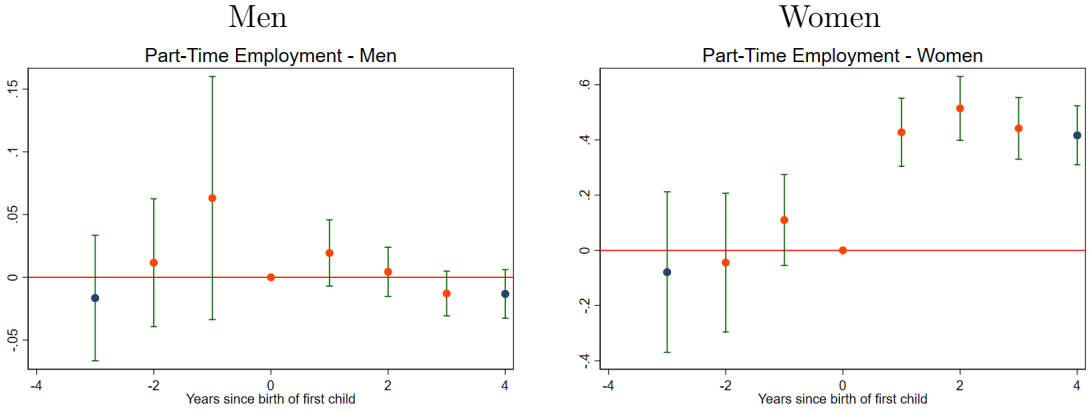
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Figure A.1: Log hourly wage by gender, working hours and firm-size bin



Note: Raw log hourly wages from GSOEP data.

Figure A.2: Childbirth and part-time jobs: event study coefficients



Note: This figure reports event-study estimates of the impact of the first childbirth on the probability of moving to a part-time schedule for fathers (left panel) and mothers (right panel). The sample includes all employed parents. Controls for age, tenure and state of residence are included. Confidence intervals at 95 percent level are obtained from worker-level cluster-robust standard errors.

Appendix

A Additional Empirical Patterns

Diverging career paths across genders after childbirth. We compare labor market outcomes of parents of children below 3 years old using the German Socio-Economic Panel

(GSOEP). Specifically, we estimate OLS regressions of labor market outcomes on binary indicators with value one if the respondent is the mother (father) of a child younger than three years old, while controlling for gender, age, tenure, number of children and fixed effects for education, state of residence, and survey year. This allows us to distinguish how the labor market outcomes of mothers of young children differ from those of the fathers of young children, or to those of other women and other men (the baseline category). As reported in Table A.1, earnings differentials between men and women are larger when considering parents of young children: mothers of young children have systematically lower earnings than other women, while fathers of young children have higher earnings than other men. The earnings differentials are mostly driven by the differentials in real hourly wages, but mothers of young children work on average 4.6 hours less than men, while there is no difference in working hours between fathers of young children and other men. Mothers of young children have a lower probability of being employed. Parents of young children work fewer days in a week, but only mothers dedicate more time to childcare and housework, while fathers of young children spend on average less time than other men (including fathers of older children) on these activities. Finally, we find that the job search of women which is already on average lower than that of men is reduced even more when their children are young.

Transition between working hours after childbirth. We have used the GSOEP data to examine transitions in working hours schedules following childbirth. Specifically, we computed the share of workers in full-time jobs who had a child in a given year and analyzed their employment status, working hours, and whether they were employed at the same firm in the subsequent year. The estimates in Table A.2 indicate that 81.3% of workers remain in full-time employment with the same employer, while 7.5% transition to full-time positions with a new employer. Additionally, 8% shift to part-time schedules within their old firm, whereas 6% move to part-time positions with a new employer. A smaller fraction, 2.6%, exits employment altogether. These figures highlight that a substantial portion of working hours adjustments occurs within the same employer, possibly reflecting the German legal framework supporting the right to part-time work after childbirth. Our model explicitly accounts for these transitions across working hours schedules.

Table A.1: Labor market differences for men and women with or without young children

Dependent variable	(1) Earnings	(2) Log real hourly wage	(3) Hours	(4) Employment	(5) Days worked per week	(6) Child care hours per week	(7) Housework hours	(8) Job search
Mother of a child ≤3 y.o.	-274.3***	-0.0759***	-4.594***	-0.029***	-0.187***	3.071***	0.372***	-0.127***
	(52.15)	(0.0277)	(0.554)	(0.007)	(0.024)	(0.167)	(0.0669)	(0.036)
Father of a child ≤3 y.o.	129.6**	0.0654***	0.507	-0.005	-0.037**	-0.824***	-0.215***	0.005
	(51.81)	(0.0243)	(0.482)	(0.006)	(0.015)	(0.129)	(0.0578)	(0.036)
Woman	-731.2***	-0.154***	-10.09***	0.002	-0.453***	0.978***	1.412***	-0.140***
	(28.52)	(0.0164)	(0.364)	(0.003)	(0.011)	(0.0837)	(0.0393)	(0.016)
Age	-7.369***	0.00149*	-0.241***	0.001***	-0.008***	-0.115***	0.0152***	-0.003***
	(1.134)	(0.000859)	(0.0185)	(0.000)	(0.001)	(0.00456)	(0.00221)	(0.001)
Household Size	-9.025	0.0170***	-1.373***	0.025***	-0.052***	0.456***	0.251***	-0.065***
	(8.096)	(0.00611)	(0.129)	(0.001)	(0.005)	(0.0381)	(0.0162)	(0.005)
Observations	15,983	15,306	13,006	12,3947	68,792	14,981	15,561	20,443
R-squared	0.316	0.166	0.262	0.064	.078	0.327	0.334	0.135

Note: Linear probability models with robust standard errors in parentheses. Controls include age, tenure, current household size, number of children and fixed effects for schooling level categories, state, survey year. *,**,*** denote statistical significance at the 10, 5 and 1 percent level.

Table A.2: Transition across working hours

	Transition rate
Full time old firm to full time old firm	0.813
Full time old firm to full time new firm	0.075
Full time old firm to part time old firm	0.080
Full time old firm to part time new firm	0.060
Full time old firm to non-employment	0.026

Note: Share of workers in full time jobs having a child at t, by employment, working hours, and employer the next year $t+1$. Own computation using GSOEP data using respondents observed in the year before and after childbirth. The estimates are robust when using a three-year lag but the sample is reduced.

B Model: Further Details

B.1 Single Households

We denote by $V^g(x,k)$ the value function of a single household with gender g where $x = (w,h,s,p)$ is the worker state vector and k is the family state. The Bellman equation of the household is

$$\begin{aligned} D^g(x,k)V^g(x,k) = u(Y^s(x,k),h) + \sum_{k'} \Phi_{kk'}\bar{V}^g(x,k') + \pi_x^+ V^g(x^+,k) + \pi_x^- V^g(x^-,k) \quad \text{(B.1)} \\ + \sum_{\hat{h}\in H_+} \kappa_{h,p}\lambda^{g,\hat{h}} \int \max\left[V^g(x,k), V^g(\hat{x},k)\right] dF^{g,\hat{h}}(\hat{w}) \\ + \mathbb{I}_{p=1}\tau \max\left[V^g(x^r,k), V^g(x^n,k), V^g(x^l,k)\right] + \delta_h V^g(x^n,k) \end{aligned}$$

where $D^g(x,k) \equiv \rho + \sum_{k'}\Phi_{kk'} + \pi_x^+ + \pi_x^- + \sum_{\hat{h}}\kappa_{h,p}\lambda^{g,\hat{h}} + \mathbb{I}_{p=0}\tau + \delta_h$. We use the same shortened notation for the continuation states of the worker as defined in the main text: $x^+ = (w,h,s^+(s),p)$ and $x^- = (w,h,s^-(s),p)$ denote skill adjustment, $\hat{x} = (\hat{w},\hat{h},s,0)$ is the continuation state after accepting a new job $(\hat{w},\hat{h})$, $x^n = (0,0,s,0)$ is non-employment, $x^r = (w,h,s,0)$ is return from parental leave at required hours, and $x^l(w,h,s,2)$ is the return from parental leave at reduced hours.

The first line of Bellman equation (B.1) includes the continuation utility after a change of the family state. As explained for couple households in the main text, the worker may quit the job (when $k \neq 2$), enter parental leave which dominates quitting when $k = 2$, may return from parental leave (at required or reduced hours) or quit the job when children grow older ($k = 3$), or enter retirement when $k = K$. Therefore,

$$\bar{V}^g(x,k) = \begin{cases} \max\left[V^g(x,k), V^g(x^p,k)\right] & \text{if } k = 2\ , \\ \max\left[V^g(x^r,k), V^g(x^l,k), V^g(x^n,k)\right] & \text{if } k = 3\ , \\ \max\left[V^g(x,k), V^g(x^n,k)\right] & \text{if } k \neq 2,3,K\ , \\ 0 & \text{if } k = K\ . \end{cases}$$

Decisions about job acceptance, quitting, or entering parental leave involve binary policy functions where we use the notation

$$\Omega^g\left(x^0;(x^i)_{i=1}^I;k\right) = \begin{cases} 1 & \text{if } V^g(x^0,k) > V^g(x^i,k) \text{ for all } i = 1,\ldots,I\ , \\ 0 & \text{else}\ , \end{cases}$$

to indicate that x^0 is preferred over any of the alternatives (x^i), $i > 0$.

B.2 Taste Shocks

In our Bellman equations, we use household discrete choices over alternatives $\boldsymbol{z} = (z^i)_{i=0}^I$ where $z^i = (x_g^i, x_{-g}^i)$ for couple households and $z^i = x^i$ for single households, with continuation utilities

$$\tilde{V}(\boldsymbol{z}, k) = \max\left(V(z^0, k), V(z^1, k), \ldots, V(z^I, k)\right) \ .$$

We also define the binary policy function indicating the choice of z^0 over the alternatives $(z^i)_{i=1}^I$:

$$\Omega\left(z^0; (z^i)_{i=1}^I; k\right) = \begin{cases} 1 & \text{if } V(z^0, k) > V(z^i, k) \text{ for all } i = 1, \ldots, I \ , \\ 0 & \text{else} \ . \end{cases}$$

In the numerical implementation of the model, we smooth out these discrete choices by adding idiosyncratic taste shocks α^i to continuation utilities where α^i is type-I extreme-value distributed with parameter σ_α. Given a realization of taste shocks $\boldsymbol{\alpha} = (\alpha^i)_{i=0}^I$, the household's continuation utility is

$$\tilde{V}_{\boldsymbol{\alpha}}(\boldsymbol{z}, k) = \max\left(V(z^0, k) + \alpha^0, V(z^1, k) + \alpha^1, \ldots, V(z^I, k) + \alpha^I\right) \ ,$$

so that the ex-ante continuation utility is

$$\tilde{V}(\boldsymbol{z}, k) = \mathbb{E}_{\boldsymbol{\alpha}} \tilde{V}_{\boldsymbol{\alpha}}(\boldsymbol{z}, k) = \frac{1}{\sigma_\alpha} \ln\left\{\sum_{i=0}^I \exp\left[\sigma_\alpha V(z^i, k)\right]\right\} \ .$$

The proportion of households choosing z^0 becomes

$$\Omega\left(z^0; (z^i)_{i=1}^I; k\right) = \frac{\exp\left[\sigma_\alpha V(z^0, k)\right]}{\sum_{i=0}^I \exp\left[\sigma_\alpha V(z^i, k)\right]} \ .$$

We choose a large value of parameter σ_α such that the modified model approximates the original model without taste shocks well and such that alternative (large) values of parameter σ_α do not alter our model calibration and results.

B.3 Separate Tax Function

The benchmark joint tax function maps annual household gross labor income y into net income

$$N(y) = \tau_0 \, y^{1-\tau_1} \ , \tag{B.2}$$

so that the household tax payment is $T(y) = y - \tau_0\, y^{1-\tau_1}$. For the separate-tax counterfactuals, let the two spouses earn annual gross labor incomes y_m and y_f, with $y = y_m + y_f$. The separate-tax net income of spouse $i \in \{m, f\}$ is

$$N^{sep}(y_i) = \tau_0^{sep}\, y_i^{1-\tau_1} \ , \tag{B.3}$$

where τ_1 is the same curvature (progressivity) parameter as in the joint tax function, and the corresponding tax payment is $T^{sep}(y_i) = y_i - \tau_0^{sep}\, y_i^{1-\tau_1}$. We choose the level parameter τ_0^{sep} so that the reference couple pays the same total tax under the separate and the joint schedules,

$$T^{sep}(y_m) + T^{sep}(y_f) = T(y_m + y_f) \ , \tag{B.4}$$

or, equivalently, so that total net income is equalized. Solving for the level parameter gives

$$\tau_0^{sep} = \frac{\tau_0\, (y_m + y_f)^{1-\tau_1}}{y_m^{1-\tau_1} + y_f^{1-\tau_1}} \ . \tag{B.5}$$

The reference couple (y_m, y_f) is defined from the middle of the benchmark distribution of household gross labor income: we keep the couple-month observations whose household income lies between the 45th and 55th percentiles, and set y_m (respectively y_f) equal to the median male (respectively female) annual gross labor income within this central band. The resulting calibrated level parameter is $\tau_0^{sep} = 44.90$, compared with $\tau_0 = 55.72$ under joint taxation.

B.4 Static Labor Supply Models

We consider static models of labor supply at the extensive and intensive margins, comparing couple and single workers. A couple worker pools income with a spouse. As in the main model, utility is logarithmic in consumption and additive in disutility from work for which we impose additional assumptions below.

Both models imply that labor supply of couple workers is more elastic. The intuition is as simple as the models themselves: in the presence of a spouse that contributes to household income, it takes a larger wage increase to incentivize the worker to take up work or to work longer hours.

B.4.1 Intensive Margin Labor Supply

The worker chooses hours $h \geq 0$ to maximize

$$u(wh) - v(h) \ ,$$

where disutility from hours is an increasing and convex function, and utility from consumption is

$$u(x) = \begin{cases} \ln(x) & , \text{ single,} \\ 2\ln\left(\dfrac{x + x_0}{2}\right) & , \text{ couple,} \end{cases}$$

and $x_0 \geq 0$ are earnings of the spouse. Differentiating the first-order condition $u'(wh) \cdot w = v'(h)$ with respect to the wage yields the slope of labor supply

$$h'(w) = \frac{u'(wh) + u''(wh) \cdot wh}{v''(h) - u''(wh) \cdot w^2} \ ,$$

so that the elasticity of labor supply is

$$\varepsilon_h(w) = \frac{h'(w)\, w}{h(w)} = \frac{1 + \varepsilon_{u'}(wh)}{\varepsilon_{v'}(h) - \varepsilon_{u'}(wh)} \ ,$$

where $\varepsilon_{u'}$ and $\varepsilon_{v'}$ are the elasticities of marginal utility u' and marginal hours disutility v'.

For the single worker, $\varepsilon_{u'} = -1$ implies that $\varepsilon_h \equiv 0$, so that labor supply is perfectly inelastic. For the couple worker,

$$\varepsilon_{u'}(wh) = -\frac{wh}{wh + x_0} \in (-1, 0) \ ,$$

for $x_0 > 0$, and so labor supply has a positive elasticity. In conclusion, labor supply of couple workers is more elastic.

B.4.2 Extensive Margin Labor Supply

An employed worker earns wage w, while a non-employed worker earns income b and enjoys utility from home hours equal to v, where v has differentiable cumulative distribution function G, which we assume to be a power function. A worker with utility from home hours v decides to work iff

$$u(w) - u(b) \geq v \ ,$$

with utility function $u(x)$ defined for couple and single workers as above.

Labor supply is $L(w) = G\big(u(w) - u(b)\big)$ with elasticity

$$\varepsilon_L(w) = \frac{L'(w)\,w}{L(w)} = \varepsilon_G \cdot \varepsilon_{u(\cdot)-u(b)}(w) \ .$$

Because G is a power function, the elasticity ε_G is constant and the same for couple and single workers. The second elasticity is

$$\varepsilon_{u(\cdot)-u(b)}(w) = \begin{cases} \dfrac{1}{\ln(w) - \ln(b)} & \text{, single,} \\ \dfrac{w}{w + x_0} \cdot \dfrac{1}{\ln(w + x_0) - \ln(b + x_0)} & \text{, couple,} \end{cases}$$

where x_0 is the wage (= earnings) of the spouse. Labor supply of couple workers is more elastic iff

$$w \cdot \ln\!\left(\frac{w}{b}\right) > (w + x_0) \cdot \ln\!\left(\frac{w + x_0}{b + x_0}\right) \ ,$$

which is true for any $x_0 > 0$ and $w > b$. To see this, differentiate the right-hand side with respect to x_0 to obtain the derivative

$$\ln\!\left(\frac{w + x_0}{b + x_0}\right) - \frac{w - b}{b + x_0} \ ,$$

which is negative (if $w > b$) because $\ln(y) < y - 1$ holds for any $0 < y \neq 1$.

C Numerical Procedures

C.1 Numerical Solution of Equilibrium

We solve the stationary equilibrium by iterating on the four wage schedules $w_{g,h}(y)$, for $g = m, f$ and $h = h^P, h^F$, and on the hours-posting probabilities $\Delta^h(y)$ of the model in Section 2. The procedure follows Fang and Shephard (2019) and Aizawa and Fang (2020), adapted to the present setting in which a firm chooses an hours requirement but can offer different wages to men and women.

1. Initialization. Discretize the firm-productivity distribution Γ and denote the productivity grid by $\mathbf{y} = (y_1, y_2, \ldots, y_{N_y})$, with probability mass γ_i at node y_i. For each grid point y_i, provide initial guesses of the four wages $w_{i,g,h} = w_{g,h}(y_i)$ and the two hours-choice probabilities $\Delta_i^h = \Delta^h(y_i)$, for $g \in \{m, f\}$ and $h \in \{h^P, h^F\}$. We refer to $\mathbf{w}_{g,h} = (w_{1,g,h}, \ldots, w_{N_y,g,h})$ as the derived wage grid for gender g and hours type h. The wage schedules are initialized to be increasing in productivity, and $\Delta_i^{h^P} + \Delta_i^{h^F} = 1$ at every productivity node.

2. Offer distributions. At a candidate iterate, construct the aggregate hours shares and the conditional offer masses as

$$\Delta^h = \sum_{i=1}^{N_y} \gamma_i \Delta_i^h \,, \qquad f_i^{g,h} = \frac{\gamma_i \Delta_i^h}{\Delta^h} \,. \tag{C.1}$$

The right-hand side is common across genders because men and women meet the same productivity distribution conditional on hours. Let $\mathbf{y}^{g,h} \equiv \{f_1^{g,h}, \ldots, f_{N_y}^{g,h}\}$ be the probability mass density of wage offers given g, h.

3. Household problem and stationary distribution. Given Δ^h and $\mathbf{y}^{g,h}$, solve the value functions of single and couple households by value-function iteration. Construct the job-acceptance, job-quitting, parental-leave, return-to-work, and hours-reduction policies, and use them to obtain the stationary household distribution and simulated household panels. All value and policy functions involving an employed household member are evaluated on the corresponding derived wage grid $\mathbf{w}_{g,h}$.

4. Labor supply. Let $L_i^{g,h}$ denote the stationary mass of efficiency units supplied to jobs at node (i, g, h). The discrete counterpart of equation (2.3) is

$$\ell_i(g,h) = \frac{L_i^{g,h}}{\Delta^h f_i^{g,h}} = \frac{L_i^{g,h}}{\gamma_i \Delta_i^h} \ . \tag{C.2}$$

Each worker contributes skill times actual hours $sH_p(h)$ to $L_i^{g,h}$. Hence, a worker on parental leave contributes zero, while a worker exercising the right to reduce hours contributes sh^P even though the contractual job has hours requirement h^F.

5. Firm problem. Given labor supply, update the wage and hours-posting policies. At productivity node i, decompose the total profit in equation (2.4) into the contributions associated with offers to each gender:

$$\pi_{i,g,h} = (\eta_{g,h} y_i - w_{i,g,h})\, \ell_i(g,h) \ , \qquad \pi_{i,h} = \sum_{g \in \{m,f\}} \pi_{i,g,h} \ . \tag{C.3}$$

To solve firms' problem, we need to first solve for optimal wages posted by the least-productive firm using the fine-grid method. For each gender-hours pair, construct the fine grid

$$\mathcal{W}_1^{g,h} = \text{grid}\left[0.1 w_{1,g,h},\ w_{1,g,h} + 0.95\,(w_{2,g,h} - w_{1,g,h})\,;\ N_w^{\text{fine}}\right] \ . \tag{C.4}$$

The lower bound is positive and the upper bound lies strictly below $w_{2,g,h}$, preserving the ordering of the first two wage nodes. For each candidate $\widetilde{w} \in \mathcal{W}_1^{g,h}$, replace only $w_{1,g,h}$, leaving the other wage nodes at their values from the current equilibrium iteration. Resolve the household problems, reconstruct the policy functions, and simulate the household panels to obtain $\ell_1(g,h;\widetilde{w})$. The updated wage policy for the least productive firm is

$$w_{1,g,h}^{+} = \arg\max_{\widetilde{w} \in \mathcal{W}_1^{g,h}} \left\{(\eta_{g,h} y_1 - \widetilde{w}) \ell_1(g,h;\widetilde{w})\right\} \ . \tag{C.5}$$

Given the updated wage for y_1, we compute the profit $\pi_{1,g,h}$.

We then use the envelope condition to update wages at the remaining productivity nodes, $i = 2, 3, \ldots, N_y$. Because labor supply is computed from a simulation of one million households, sampling noise makes the resulting labor-supply schedule irregular. We therefore smooth the schedule before applying the envelope update.[10] Specifically, for each gender-

[10]The high-dimensional state space makes it computationally infeasible to solve for the stationary distribution by direct matrix inversion or by iterating on the stationary stock-flow equations. We therefore approximate the stationary distribution by simulation.

hours pair, we fit the power function with parameters $a_{g,h}$, $b_{g,h}$, and $c_{g,h}$

$$\widehat{\ell}(g,h,w) = a_{g,h} w^{b_{g,h}} + c_{g,h} \tag{C.6}$$

which has the advantage of a monotone derivative.

In continuous productivity, the counterpart of equation (C.3) satisfies

$$\frac{d\pi_{g,h}(y)}{dy} = \eta_{g,h}\ell(g,h,w_{g,h}(y)) \tag{C.7}$$

and therefore

$$\pi_{g,h}(y) = \pi_{g,h}(y_1) + \int_{y_1}^{y} \eta_{g,h}\ell(g,h,w_{g,h}(x))\,dx \ . \tag{C.8}$$

For $i \geq 2$, approximate the integral by the trapezoidal rule:

$$\pi_{i,g,h} = \pi_{1,g,h} + \sum_{r=2}^{i} (y_r - y_{r-1})\eta_{g,h} \frac{\widehat{\ell}_{r-1}(g,h) + \widehat{\ell}_r(g,h)}{2} \ , \tag{C.9}$$

where $\widehat{\ell}_i(g,h) = \widehat{\ell}(g,h,w_{i,g,h})$. Solving the discrete profit identity equation (C.3) for the wage gives the update

$$w^{+}_{i,g,h} = \eta_{g,h} y_i - \frac{\pi_{i,g,h}}{\widehat{\ell}_i(g,h)} \ . \tag{C.10}$$

This produces a separate wage update for men and women for each hours type. After summing the gender-specific profit contributions, update the hours-posting probabilities using equation (2.5).

6. Iteration and convergence. Apply damping to the wage updates and repeat steps 2–5 until both the maximum relative change in the four wage schedules and the maximum absolute change in the two hours-posting probabilities fall below their respective tolerances.

C.2 Numerical Implementation of Demand-Side Calibration

Step 1 calibration results in discrete wage distributions $\{w_{i,g,h}, f_i^{g,h}\}$ and labor supply $\{\ell_i(g,h)\}$. In implementing Step 2 of the two-step calibration procedure described in Section 3.3.2, we first recover the productivity grid using the discrete counterpart of equation (3.3). As discussed in Section C.1, we fit the simulation-based labor supply function using equation (C.6). At wage node $w_{i,g,h}$, its derivative is $\partial_w \widehat{\ell}(g,h,w_{i,g,h}) = a_{g,h} b_{g,h} w_{i,g,h}^{b_{g,h}-1}$. Productivity at the

wage nodes is therefore

$$y_{i,m,h} = w_{i,m,h} + \frac{\widehat{\ell}(m,h,w_{i,m,h})}{\partial_w \widehat{\ell}(m,h,w_{i,m,h})} ,$$
$$y_{i,f,h} = \frac{1}{\eta^h}\left[w_{i,f,h} + \frac{\widehat{\ell}(f,h,w_{i,f,h})}{\partial_w \widehat{\ell}(f,h,w_{i,f,h})} \right] . \tag{C.11}$$

equation (C.11) may generate four different implied productivity grids, one for each gender-hours pair. The implementation constructs a common evenly spaced grid $\mathbf{y}$ whose lower and upper bounds are the averages of the four implied lower and upper bounds, respectively. For each $y_i \in \mathbf{y}$, the wages $w_{g,h}(y_i)$ are obtained by PCHIP, a shape-preserving piecewise-cubic interpolation, with extrapolation where necessary.

At common productivity node i, total profit from posting hours h is

$$\pi_{i,h} = (y_i - w_{m,h}(y_i))\,\widehat{\ell}(m,h,w_{m,h}(y_i)) + \left(\eta^h y_i - w_{f,h}(y_i)\right)\widehat{\ell}(f,h,w_{f,h}(y_i)) . \tag{C.12}$$

The theoretical hours-posting probability is

$$\Delta^h_{\text{th},i} = \frac{\exp\{[\pi_{i,h} - \bar{\varepsilon}_h]/\sigma_\varepsilon\}}{\sum_{\tilde{h}\in\{h^P,h^F\}} \exp\{[\pi_{i,\tilde{h}} - \bar{\varepsilon}_{\tilde{h}}]/\sigma_\varepsilon\}} . \tag{C.13}$$

To construct the observed hours-posting decision, we interpolate $\{f_i^{g,h}\}$ at the wages associated with the common productivity nodes $\{w_{g,h}(y_i)\}$ and normalize the interpolated distribution such that $\sum_i f_{g,h}(w_{g,h}(y_i)) = 1$.

The discrete counterpart of the conditional productivity density (equation (3.5)) is $q_i^{g,h} \equiv f_{g,h}(w_{g,h}(y_i))$. Because the separately estimated gender-specific offer distributions need not satisfy the exact common-productivity restriction, the implementation combines them into the unconditional joint mass

$$M_i^h = \Delta^h \frac{1}{2} \sum_{g\in\{m,f\}} q_i^{g,h} . \tag{C.14}$$

The observed hours-posting decision and the productivity mass at node i are

$$\Delta^h_{\text{ob},i} = \frac{M_i^h}{\sum_{\tilde{h}} M_i^{\tilde{h}}} , \qquad \gamma_i = \frac{\sum_h M_i^h}{\sum_r \sum_h M_r^h} . \tag{C.15}$$

D Estimation of Moments

We obtain all statistics for model calibration from the German Socio-Economic Panel (GSOEP), a representative household survey conducted by DIW Berlin since 1984. Each year, around 30,000 individuals from 15,000 households are surveyed, providing information on household composition, labor market status, employment history, working hours, earnings, transfers, and other income sources, along with various personal and household characteristics. We use GSOEP wave 37 and data from the survey years 2007 to 2019. We do not include earlier years, as the parental leave reforms central to our benchmark model took effect in January 2007. Later years are excluded because the COVID-19 pandemic has impacted the labor market behavior of workers in unprecedented ways.

Sample and definitions. We define working-age individuals as those who are 20 years or older. We limit our analysis to households where the head is between 20 and 65 years old, categorizing them into the following household types:

- *Single household - Man*: The household head is a man, with no other working-age individuals present.
- *Single household - Woman*: The household head is a woman, with no other working-age individuals present.
- *Couple household*: A married couple (a man and a woman) is present, both aged 20-65, with no other working-age individuals in the household.

We exclude all households that do not satisfy these definitions, including cohabiting or same-sex couples, singles or couples with working-age children, and households with one spouse over 66 and the other below 65.[11] For all three household types, we define the following demographic groups:

- *Childless fertile*: In single households, the head is under 40. In couple households, the woman is under 40.
- *Young kids*: The household has children, with the youngest being at most 7 years old.
- *Old kids*: The household has children, with the youngest being 8 years or older.

[11] If one individual identifies as married and cohabiting, but other members aged 20-65 have missing marital status, we impute the closest-aged individual as the spouse. We exclude households with missing age or gender for any member, as this would prevent accurate identification of household heads, children, and demographic groups.

- *Childless infertile*: The household has no co-habitating children (if there were, they must have moved out). In single households, the head is 40 or older. In couple households, the woman is 40 or older.

The final sample includes 570,799 observations, corresponding to 110,243 unique workers and 40,655 unique households.

Employment and wages. Workers are classified as full-time employed if their weekly working hours (*pgtatzeit*) exceed 30, part-time employed if their weekly working hours are between 8 and 30, or non-employed otherwise. On average, full-time employees work 43 hours each week, while part-time employees work about 20 hours per week (Table 3.2).

We calculate the hourly wage of employed workers as current gross labor income (*pglabgro*) divided by monthly hours (4.3 times the weekly working hours). We deflate all wages using the OECD Consumer Price Index for Germany with a base year of 2015 as reported in the FRED database. We compute the distribution of log weekly wages based on gender and work schedule. On average, women earn 92% of men's hourly wages when working full time (2.65 vs 2.87 log euro per hour). However, average hourly wages are nearly the same between genders in part-time roles. Men working part-time earn about 91% of those working full-time, while part-time women earn approximately 98% of full-time women's wages. Additionally, we examine wage distribution by gender at the start of a worker's career, focusing on employees under 28 years old in full-time positions. The average log wage during the first job spell, compared to the average log wage over the course of a working life, is 85% for men and 92% for women.

Taxes and Transfers. We estimate the parameters of the tax functions for single and couple households (equation (3.1) and equation (3.2), respectively), following the approach of Kaas et al. (2021). Specifically, we use CPI-adjusted income data from the GSOEP's *pequiv* module. We log-transform each equation and estimate it using Ordinary Least Squares for the combined data set, with separate estimations for each household type and demographic group. We define net annual household income as gross annual income minus taxes plus transfers (including child allowances). We then regress net log income on gross log income separately for single and couple households (pooling genders), while controlling for year fixed effects. The estimated constant and slope are used as parameters for the tax functions of the two household types. The parameter τ_0, calculated as 1 minus the constant, reflects the net tax revenue and represents the average level of taxation. The parameter τ_1, derived as the exponential of the slope parameter, determines the progressivity of the tax and transfer system. When $\tau_1 > 0$, marginal tax rates exceed average tax rates, indicating a progressive tax system. In contrast, negative values of τ_1 indicate a regressive system, while a value of

Table D.1: Tax function by household type and demographic group

	Dependent variable: Log net income			
Sample:	Couple Households			
Demographic group:	1	2	3	4
Log gross income	0.766***	0.436***	0.565***	0.612***
	(0.020)	(0.012)	(0.016)	(0.049)
Constant	2.467***	6.169***	4.754***	4.157***
	(0.223)	(0.134)	(0.180)	(0.540)
R-squared	0.949	0.718	0.825	0.793
N. observations	963	27,127	15,669	1,792
Sample:	Single Households			
Demographic group:	1	2	3	4
Log gross income	0.539***	0.506***	0.512***	0.576***
	(0.013)	(0.013)	(0.013)	(0.012)
Constant	4.687***	5.070***	5.004***	4.254***
	(0.139)	(0.134)	(0.132)	(0.121)
R-squared	0.686	0.647	0.656	0.693
N. observations	17,971	20,453	24,400	22,395

Note: This table reports the estimates of the regression of log net income on log gross income, both computed excluding the child and parental allowance. The first and the second columns report the estimates for the sample of couple and single households, respectively. Standard errors are reported in parentheses. *, **,*** denote statistical significance at the 10, 5, 1 percent level.

zero implies a flat tax rate.

The estimated coefficients are reported in Table D.1, and the implied tax function parameters are presented in Table 3.2. The large positive estimate of τ_1 suggests that the German tax and transfer system is highly redistributive.

E Additional Model Results

E.1 Further Model Statistics

Table E.1 shows the distribution of workers by types of employment, including parental leave and reduced hours. Table E.2 shows the male share in part-time and full-time employment.

Table E.1: Detailed employment distribution

Gender	k	NE	PL	PT	FT reduced	FT
Male	1	0.139	0.000	0.113	0.000	0.748
Male	2	0.034	0.014	0.077	0.010	0.865
Male	3	0.053	0.000	0.073	0.006	0.869
Male	4	0.240	0.000	0.077	0.001	0.682
Female	1	0.131	0.000	0.177	0.000	0.691
Female	2	0.328	0.094	0.189	0.297	0.093
Female	3	0.198	0.000	0.245	0.294	0.263
Female	4	0.292	0.000	0.147	0.094	0.467

Note: Columns show the share in each of the five detailed employment states, conditional on gender and k. NE is non-employment (excluding those on parental leave); PL is parental leave; PT is part-time contract (not on parental leave, not on full-time contract with reduced hours); FT reduced is full-time contracts with reduced hours; and FT is full-time contracts without reduced hours.

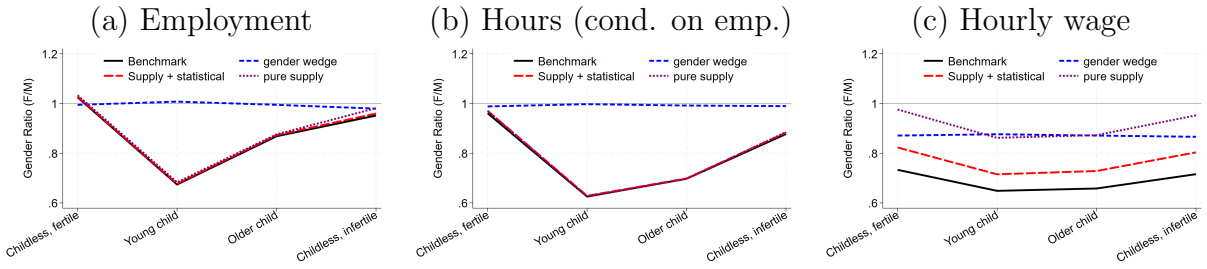
Table E.2: Share of men in part-time and full-time employment

	Part-time	Full-time
All	0.175	0.696
Childless, fertile	0.361	0.516
Young child	0.132	0.982
Older child	0.101	0.846
Childless, infertile	0.208	0.621

Note: The table shows the share of men in each type of employment.

E.2 Counterfactual Experiments

Figure E.1: Gender gap over the life cycle: employment, hours and wages



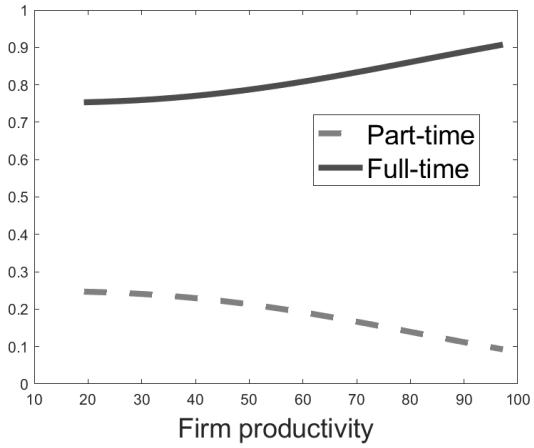
Note: Female-to-male ratio (in %) in employment, hours (conditional on employment) and hourly wages across the four demographic states, in the benchmark economy and the three counterfactuals of Table 4.1.

Figure E.2: Drivers of the gender earnings gap: supply factors



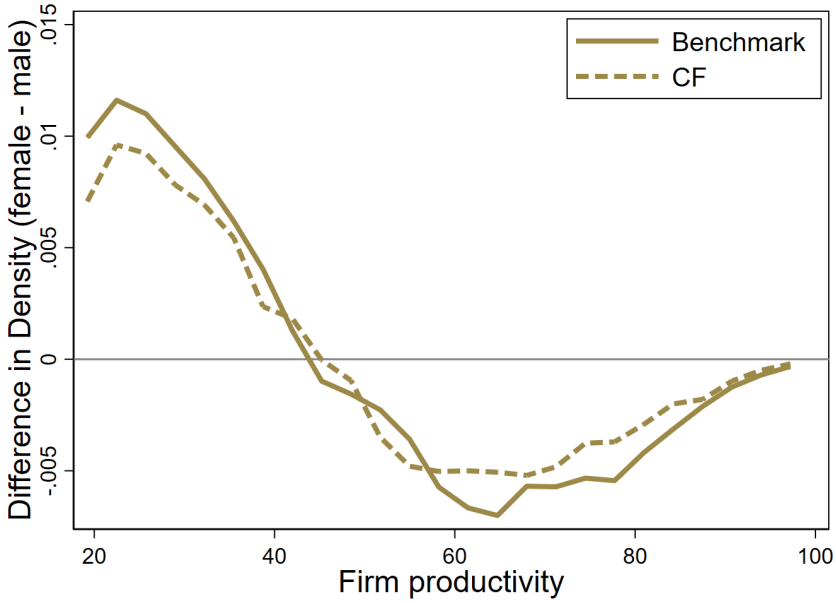
Note: Gender earnings gap (in %), decomposed into employment, hours and hourly wages, for the benchmark economy, the three counterfactuals of Table 4.1, and three additional counterfactuals that switch off gender differences in job search, initial skills and preferences one at a time.

Figure E.3: Probability of posting PT and FT jobs by firm productivity



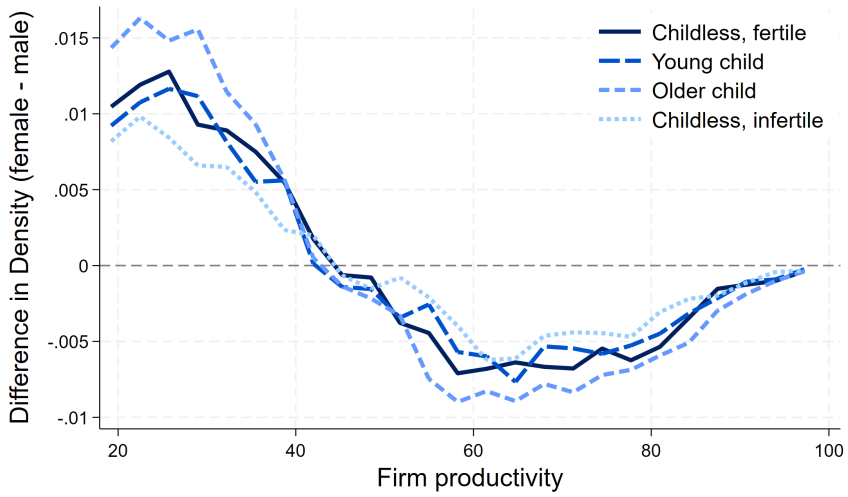
Note: Based on the policy function of firms, $\Delta(h;p)$, from the benchmark economy.

Figure E.4: Density difference (female minus male) across firm productivity: benchmark and uniform-Δ counterfactual



Note: Based on simulated data from the benchmark and counterfactual uniform-Δ economy. In the counterfactual economy, we set $\Delta(h;p)$ at the average value across firms.

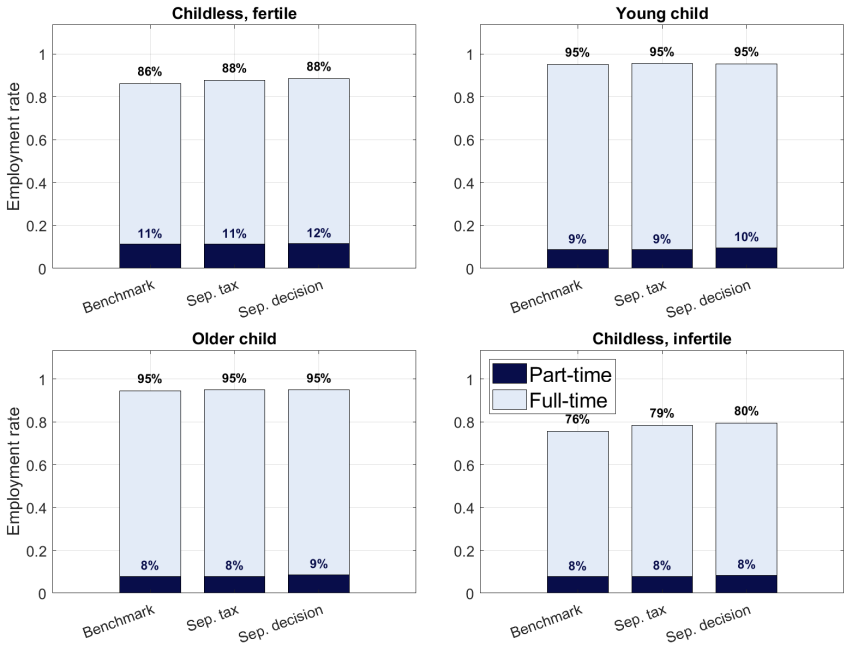
Figure E.5: Gender difference in employment across firm productivity



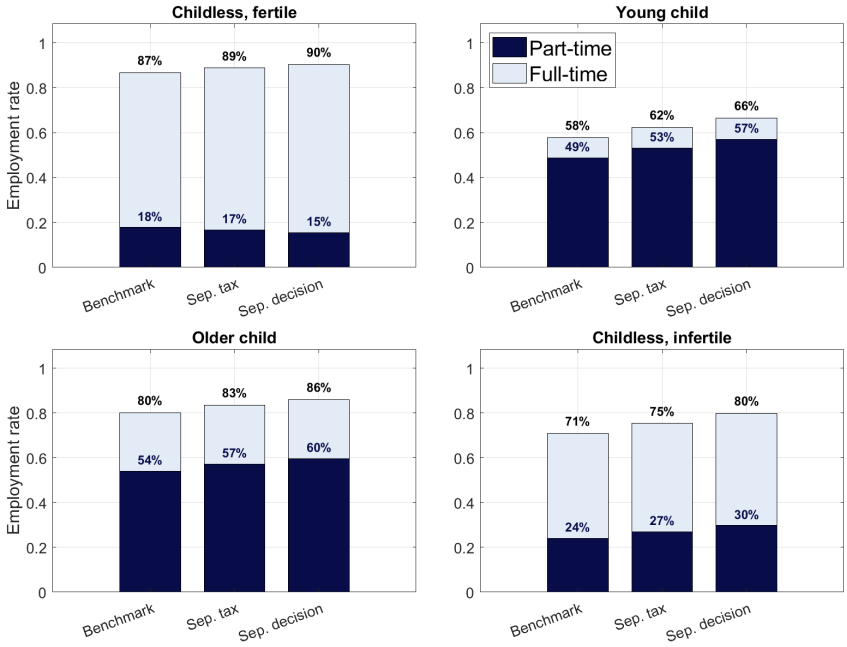
Note: Difference between the female and male employment densities (female minus male) across firm productivity, by demographic state, based on simulated data from the benchmark economy. Negative (positive) values indicate that women are less (more) frequently employed than men in firms of a given productivity.

Figure E.6: Employment distribution across the joint, separate-taxation and separate-decision economies

(a) Men



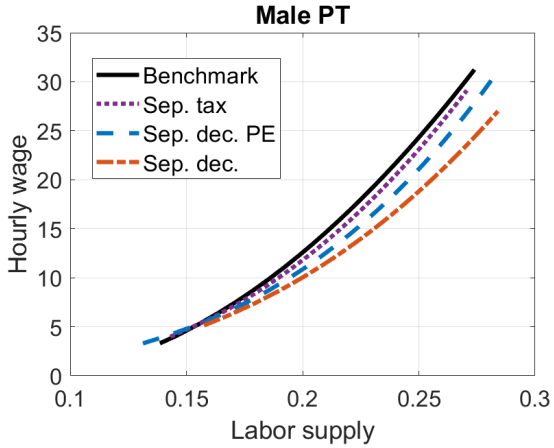
(b) Women



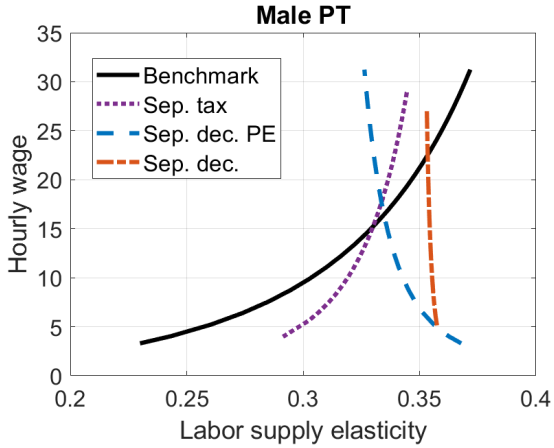
Note: Employment rates and the part-time/full-time split by gender and demographic state, in the benchmark (joint-decision), separate-taxation and separate-decision economies.

Figure E.7: Male labor supply and elasticity

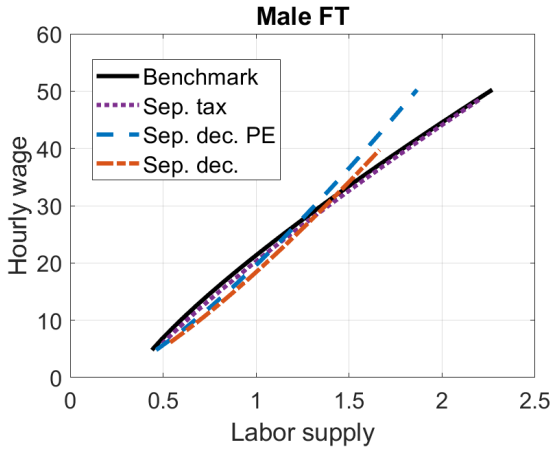
(a) PT labor supply



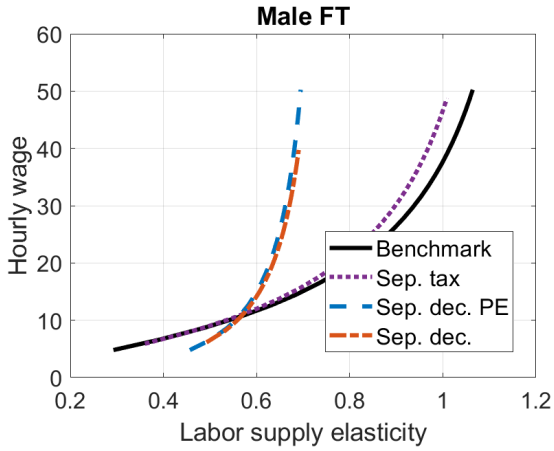
(b) PT elasticity



(c) FT labor supply



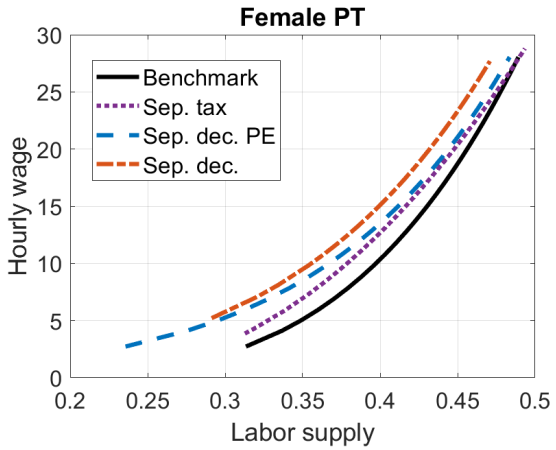
(d) FT elasticity



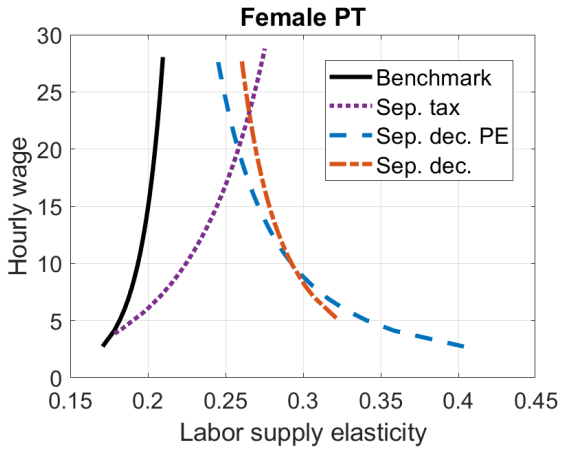
Note: Male labor supply schedules (left) and their implied wage elasticities (right), for part-time (top) and full-time (bottom) jobs, in the joint benchmark, the separate-decision economy in partial and general equilibrium, and the separate-taxation economy.

Figure E.8: Female labor supply and elasticity

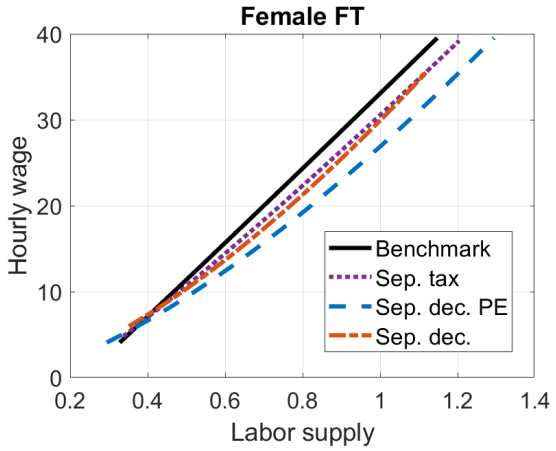
(a) PT labor supply



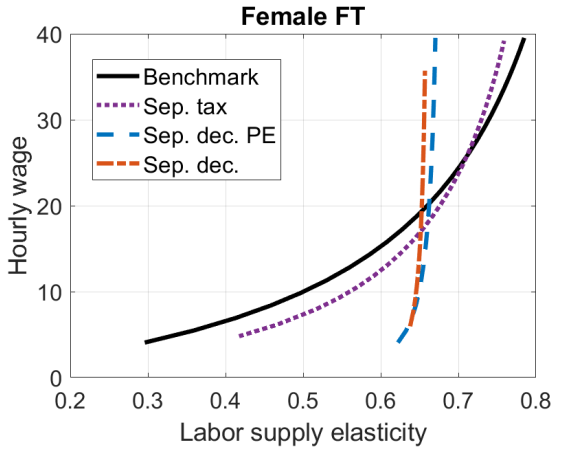
(b) PT elasticity



(c) FT labor supply



(d) FT elasticity



Note: Female labor supply schedules (left) and their implied wage elasticities (right), for part-time (top) and full-time (bottom) jobs, in the joint benchmark, the separate-decision economy in partial and general equilibrium, and the separate-taxation economy.

E.3 Source Decomposition of the Gender Earnings Gap

We decompose the gender earnings gap into contributions from the firms' gender wedge, statistical discrimination and pure supply, following the Shapley-Owen-Shorrocks approach (Audoly et al., 2025). Let f_e^1 denote the gender earnings gap in economy e, measured as the female-to-male earnings ratio minus one, so that negative values indicate lower female earnings. We set the gender-equity gap, in which all gender differences are eliminated, to $f_e^0 = 0$. Let f_e^G denote the gap in the gender-wedge counterfactual, f_e^{SD} the gap in the supply-plus-statistical-discrimination counterfactual, f_e^S the gap in the pure-supply counterfactual, and f_e^{SG} the gap in the pure-supply-plus-gender-wedge counterfactual. We apply the decomposition separately to each economy e, using the counterfactuals computed within that economy.

We first group supply and statistical discrimination together, so that $f_e^1 - f_e^0 = C_{G,e} + C_{SD,e}$, where

$$C_{G,e} = \tfrac{1}{2}\left(f_e^G - f_e^0\right) + \tfrac{1}{2}\left(f_e^1 - f_e^{SD}\right) \ , \tag{E.1}$$

$$C_{SD,e} = \tfrac{1}{2}\left(f_e^{SD} - f_e^0\right) + \tfrac{1}{2}\left(f_e^1 - f_e^G\right) \ . \tag{E.2}$$

We then split the joint supply and statistical-discrimination term into a pure-supply and a statistical-discrimination component,

$$C_{S,e} = \tfrac{1}{2}\left(f_e^S - f_e^0\right) + \tfrac{1}{2}\left(f_e^{SG} - f_e^G\right) \ , \tag{E.3}$$

$$C_{D,e} = \tfrac{1}{2}\left(f_e^{SD} - f_e^S\right) + \tfrac{1}{2}\left(f_e^1 - f_e^{SG}\right) \ , \tag{E.4}$$

so that the decomposition is $f_e^1 - f_e^0 = C_{G,e} + C_{S,e} + C_{D,e}$. Component shares are measured relative to the raw benchmark gap $f_e^1 - f_e^0$. Table E.3 reports the resulting levels and shares.

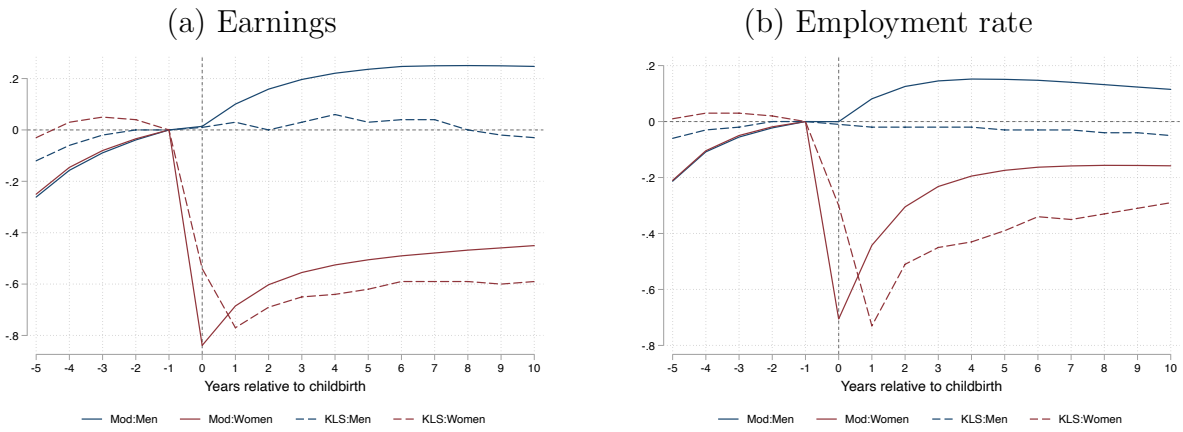
Table E.3: Source decomposition of the gender earnings gap

	Joint decision		Separate tax		Separate decision	
	Level	%	Level	%	Level	%
Gender wedge	-10.6	21.1	-7.7	18.9	-6.2	19.1
Supply + statistical	-39.6	78.9	-33.0	81.1	-26.0	80.9
Statistical	-9.6	19.2	-4.4	10.9	-0.5	1.5
Pure supply	-30.0	59.7	-28.6	70.2	-25.6	79.4
Total gender gap	-50.2	100.0	-40.7	100.0	-32.2	100.0

Note: Shapley-Owen-Shorrocks decomposition of the gender earnings gap (female-to-male earnings ratio minus one, in %) into the contributions of the firms' gender wedge, statistical discrimination and pure supply, for the benchmark (joint-decision), separate-taxation and separate-decision economies. "Supply + statistical" is the intermediate term $C_{SD,e} = C_{S,e} + C_{D,e}$.

E.4 Child Penalty

Figure E.9: Child penalty estimates around childbirth



Note: The figure compares the child penalty in earnings (panel a) and employment rate (panel b) around the birth of the first child in the model and in the data by gender. For the model, the child penalty is constructed from a simulated panel as the average of the reference variable for women and men normalized to the year before childbirth. For the data, the estimated child penalties for women and men using the event-study methodology of Kleven et al. (2019b) (KLS) are reported, with event time normalized to the year before the first birth ($t = -1$).